



Grade 8 Statewide Science Assessment

Spring 2024 Released Test Items

1	Periodic Table of the Elements Atomic Number — 1 Element Name — Hydrogen Element Symbol — H Average Atomic Mass — 1.01 Metal Metalloid (semimetals) Non-metals																18
1 H Hydrogen 1.01	2	13	14	15	16	17											2
3 Li Lithium 6.94	4 Be Beryllium 9.01	5 B Boron 10.81	6 C Carbon 12.01	7 N Nitrogen 14.01	8 O Oxygen 16.00	9 F Fluorine 19.00	10 Ne Neon 20.18										
11 Na Sodium 22.99	12 Mg Magnesium 24.30	13 Al Aluminum 26.98	14 Si Silicon 28.09	15 P Phosphorus 30.97	16 S Sulfur 32.06	17 Cl Chlorine 35.45	18 Ar Argon 39.95										
19 K Potassium 39.10	20 Ca Calcium 40.08	21 Sc Scandium 44.96	22 Ti Titanium 47.87	23 V Vanadium 50.94	24 Cr Chromium 51.996	25 Mn Manganese 54.94	26 Fe Iron 55.85	27 Co Cobalt 58.93	28 Ni Nickel 58.69	29 Cu Copper 63.55	30 Zn Zinc 65.38	31 Ga Gallium 69.72	32 Ge Germanium 72.63	33 As Arsenic 74.92	34 Se Selenium 78.97	35 Br Bromine 79.90	36 Kr Krypton 83.80
37 Rb Rubidium 85.47	38 Sr Strontium 87.62	39 Y Yttrium 88.91	40 Zr Zirconium 91.22	41 Nb Niobium 92.91	42 Mo Molybdenum 95.95	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.90	54 Xe Xenon 131.29
55 Cs Cesium 132.91	56 Ba Barium 137.33	57-71	72 Hf Hafnium 178.49	73 Ta Tantalum 180.95	74 W Tungsten 183.84	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.21	83 Bi Bismuth 208.98	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)
87 Fr Francium (223)	88 Ra Radium (226)	89-103	104 Rf Rutherfordium (267)	105 Db Dubnium (268)	106 Sg Seaborgium (269)	107 Bh Bohrium (270)	108 Hs Hassium (269)	109 Mt Meitnerium (278)	110 Ds Darmstadtium (281)	111 Rg Roentgenium (282)	112 Cn Copernicium (285)	113 Nh Nihonium (286)	114 Fl Flerovium (289)	115 Mc Moscovium (289)	116 Lv Livermorium (293)	117 Ts Tennessine (294)	118 Og Oganesson (294)

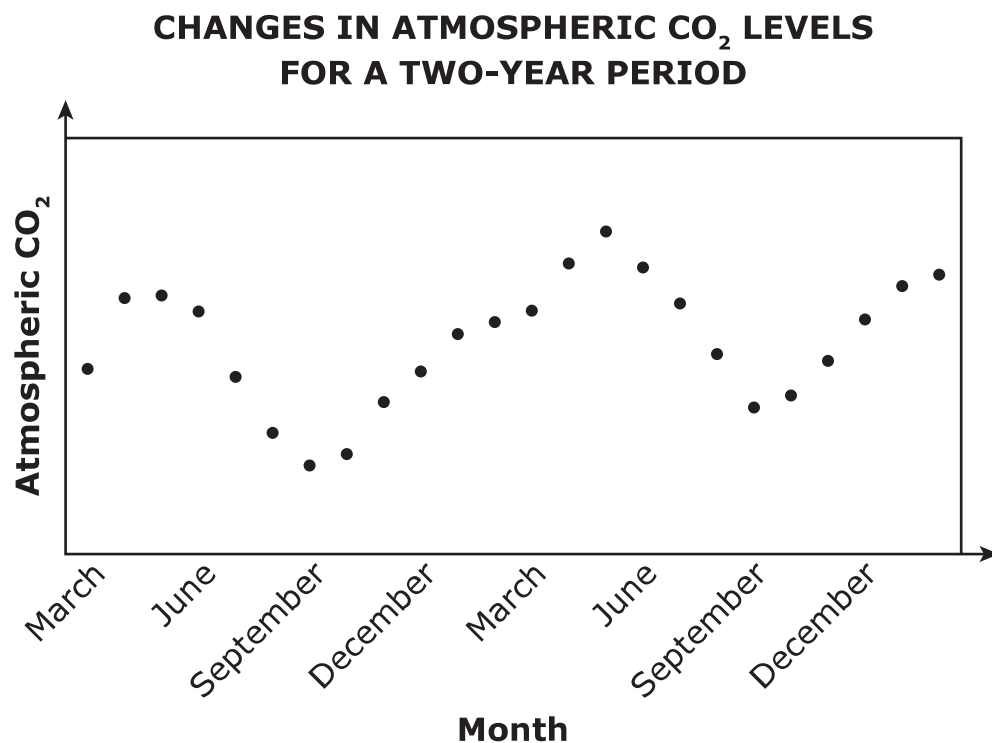


57 La Lanthanum 138.91	58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.05	71 Lu Lutetium 174.97
89 Ac Actinium (227)	90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (259)	103 Lr Lawrencium (266)

- 1.** Janice is learning about compounds. Which of the following **best** describes a compound?
- Ⓐ a substance made up purely of one element
 - Ⓑ a group of substances that are either acids or bases
 - Ⓒ a group of substances that can be separated by a physical change
 - Ⓓ a substance made up of two or more elements that are chemically combined

2. Scientific theories are developed over time in all areas of science. Which statement **best** describes scientific theories?
- Ⓐ They are based on scientific laws that came first.
 - Ⓑ They are pieces of scientific evidence that support a hypothesis.
 - Ⓒ They are explanations of scientific events that may change based on continuous research.
 - Ⓓ They are predictions of events that usually remain constant because scientific ideas rarely change.

3. Many interactions occur between the biosphere and the atmosphere. One of these interactions results in changes in atmospheric carbon dioxide (CO_2) levels. The graph below shows these changes over a two-year period.

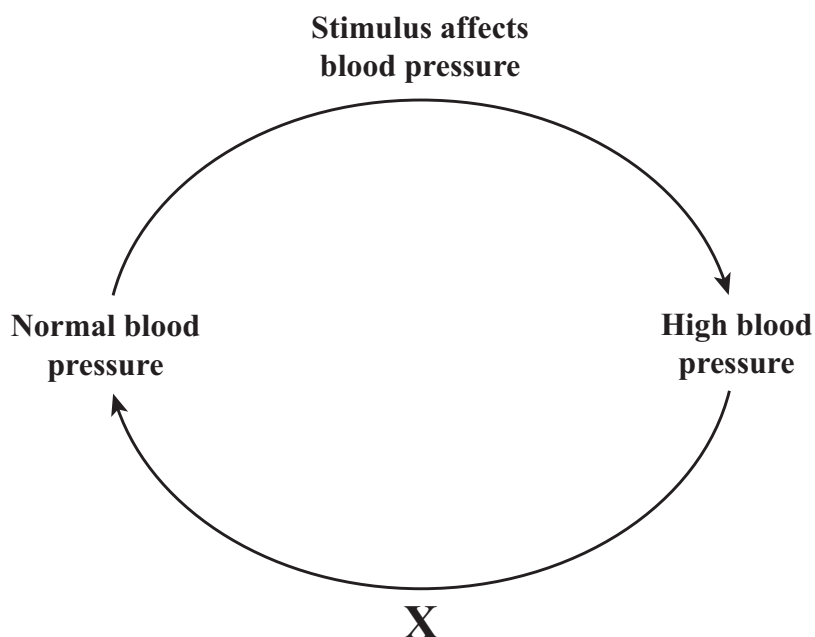


Which of the following changes in the biosphere **most likely** caused these changes in the atmospheric CO_2 levels?

- Ⓐ Streams froze in the winter and melted in the spring.
- Ⓑ Plants grew leaves in the spring and lost leaves in the fall.
- Ⓒ Offspring were born in the summer and matured in the fall.
- Ⓓ Soil dried out in the summer and absorbed moisture in the winter.

4. Jordan adds some soil and plants to a large glass jar to build a terrarium. He waters the plants and then adds two snails, two crickets, and an earthworm to the jar. Jordan then places the lid tightly on the jar and sets it on a sunny windowsill. Based on the law of conservation of mass, which of the following is expected to occur inside Jordan's terrarium?
- Ⓐ The mass will increase as the plants grow and decrease when the snails eat the plants.
 - Ⓑ The mass will increase as the earthworm digests the soil.
 - Ⓒ The mass will decrease when the crickets die.
 - Ⓓ The mass will neither increase nor decrease.

5. Human body systems interact to keep blood pressure within a normal range. The diagram below represents this interaction.



Which of the following **best** describes what is occurring in part X of the diagram?

- Ⓐ Respiration remains high as other body systems work with the circulatory system.
- Ⓑ Blood pressure is regulated as the circulatory system shuts down other body systems.
- Ⓒ Blood pressure remains high as the circulatory system shuts down other body systems.
- Ⓓ Homeostasis is maintained as the other body systems work with the circulatory system.

6. The table below shows the range of frequencies, in Hertz (Hz), for part of the electromagnetic spectrum.

Region	Frequency (Hz)
Infrared	$3.0 \times 10^{12} - 4.3 \times 10^{14}$
Ultraviolet	$7.5 \times 10^{14} - 3.0 \times 10^{17}$
X-rays	$3.0 \times 10^{17} - 3.0 \times 10^{19}$
Gamma rays	$> 3.0 \times 10^{19}$

Which of the following types of radiation has the longest wavelength?

- Ⓐ Infrared
- Ⓑ Ultraviolet
- Ⓒ X-rays
- Ⓓ Gamma rays

7. An astronomer observes a sunspot. Which of the following describes this feature of the Sun?
- Ⓐ a location where heat rises to the Sun's surface
 - Ⓑ a location that produces most of the Sun's heat energy
 - Ⓒ a location that is cooler than other parts of the Sun's surface
 - Ⓓ a location where energy and matter leave the Sun's atmosphere

Grade 8 Science

8. Ricky wants to finish constructing a chart that helps classify two organisms. The chart is shown below.

Question	Organism X	Organism Y
1. Does it require nutrients to survive?		
2. Does it have DNA?		
3. Do its cells lack cell walls?		
4. Can it reproduce?		

Accurately answering which question would **best** help distinguish these organisms as members of the animal kingdom rather than the plant or fungi kingdom?

- Ⓐ 1
- Ⓑ 2
- Ⓒ 3
- Ⓓ 4

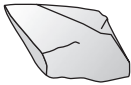
- 9.** Atoms are made up of subatomic particles, including neutrons. Which of the following describes a neutron?
- Ⓐ has a greater volume than the electron cloud
 - Ⓑ has more mass than the nucleus
 - Ⓒ has no charge
 - Ⓓ has no mass

10. Two groups of students hypothesized that rocks on Earth's surface are damaged by acid rain.

- Group X tested the hypothesis by examining properties of some rocks and pouring a weak acid, vinegar, on each of the rocks.
- Group Y later repeated the same procedure with a different group of rocks.

The rocks used for the investigations and the results from each student group are shown below.

ROCKS TESTED

Group	Rock 1	Rock 2	Rock 3	Rock 4
X	 Calcite	 Dolostone	 Shale	 Limestone
Y	 Calcite	 Dolostone	 Dolostone	 Calcite

OBSERVATIONS OF THE ROCKS

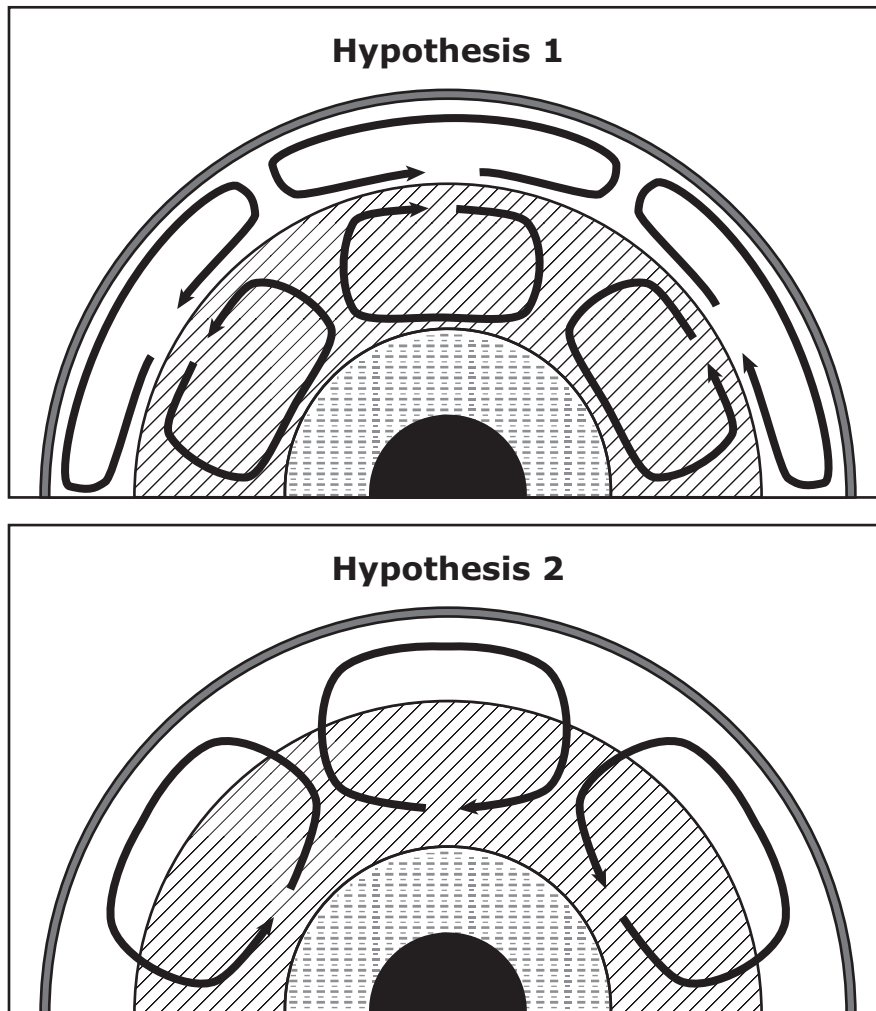
Group	Number of Rocks Used	Average Mass of Rocks Used (grams)	Texture of Rocks	Reaction to Vinegar
X	4	46	3 rocks smooth 1 rock rough	3 rocks bubbled
Y	4	42	2 rocks smooth 2 rocks rough	4 rocks bubbled

Which of the following **best** describes an advantage Group X has over Group Y in supporting the hypothesis?

- Ⓐ Group X tested the rocks before Group Y did.
- Ⓑ Group X tested more rocks that are smoother.
- Ⓒ Group X tested rocks with a greater average mass.
- Ⓓ Group X tested a greater variety of rocks than Group Y did.

-
- 11.** Mandy applies the brakes on her bicycle and stops quickly to avoid hitting a basketball that bounces into the street. Which of the following **best** explains what happens to the bicycle's kinetic energy?
- Ⓐ It increases because the tires make a screeching noise.
 - Ⓑ It is destroyed because Mandy stops the bicycle quickly.
 - Ⓒ It remains the same because Mandy is the source of energy.
 - Ⓓ It is transformed to heat because the tires are slowed by friction.

12. The diagram shows two hypotheses for the same process that occurs beneath Earth's surface.



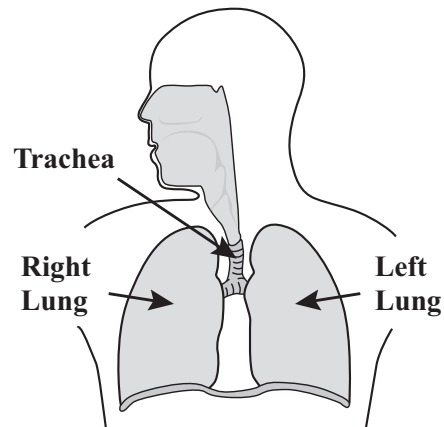
Which of the following processes is represented by these hypotheses?

- Ⓐ convection current movement in Earth's mantle
- Ⓑ plate movements in Earth's outer core
- Ⓒ layer separation in Earth's inner core
- Ⓓ rock formation in Earth's lithosphere

13. North Atlantic right whales live in the ocean near the east coast of the United States and Canada. Fewer than 500 of these whales exist. Which of the following would contribute to the extinction of the North Atlantic right whales?

- Ⓐ Species that compete with right whales move to new habitats.
- Ⓑ Humans develop breeding programs for right whales in aquariums.
- Ⓒ The environment changes faster than right whale populations can adapt.
- Ⓓ New technologies prevent right whales from being caught in fishing gear.

14. The diagram below identifies three structures found in the human body.



Which level of organization describes each of these three structures?

- Ⓐ cell
- Ⓑ organ
- Ⓒ organism
- Ⓓ tissue

15. A space probe lands on a planet that has half the pull of gravity as Earth. What effect does this have on the space probe's weight and mass?

- Ⓐ Its weight is reduced to half its weight on Earth, and its mass remains the same as on Earth.
- Ⓑ Its weight remains the same as on Earth, and its mass is reduced to half its mass on Earth.
- Ⓒ Both its weight and mass are half of what they are on Earth.
- Ⓓ Both its weight and mass remain the same as on Earth.

- 16.** Cells experience similar processes. One of these processes involves controlling cellular wastes. Why do cells undergo processes that control cellular wastes?
- Ⓐ All cells must release cellular wastes to maintain homeostasis.
 - Ⓑ Plant cells eliminate cellular wastes to regulate the rate of cell reproduction.
 - Ⓒ Animal cells extract energy from cellular wastes to perform cellular functions.
 - Ⓓ All cells must release cellular wastes to provide a means of maintaining cell size.

17. Nancy observes the Moon from the same location outside each night for one week. She notices that when it first appears each night, the Moon is a little farther east than it was the night before. Which of the following is one explanation for her observation?

- Ⓐ The Moon's speed of rotation changes as it revolves around Earth.
- Ⓑ The Moon revolves around Earth as Earth revolves around the Sun.
- Ⓒ The Moon and Earth take the same amount of time to complete one rotation.
- Ⓓ The Moon completes one rotation each time Earth completes one revolution.

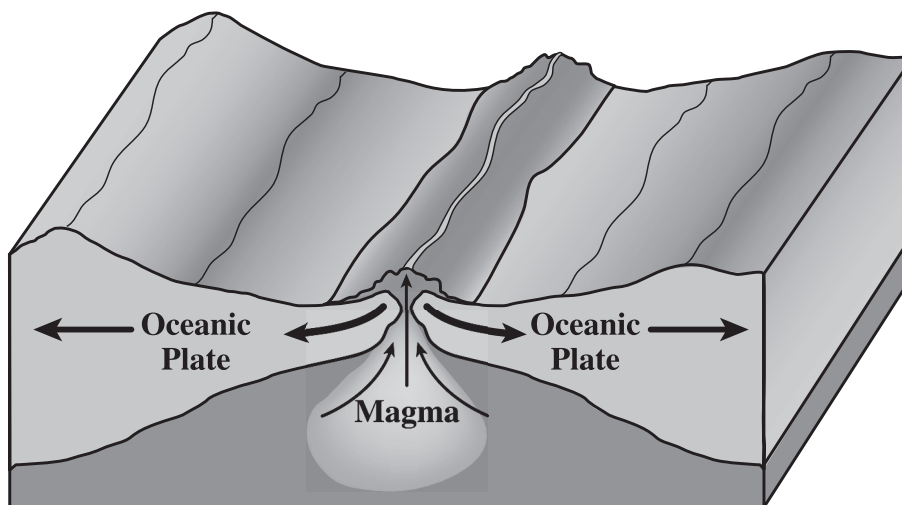
- 18.** Brittany is investigating plant growth. She completes the following activities in class.

Activity 1	Activity 2
1. Plant three identical seeds in each of four cups containing the same amount and type of soil. 2. Add different amounts of the same fertilizer to three of the cups and 50 milliliters (mL) of water to all of the cups. 3. When the seeds sprout, measure the height of the plants in each cup. 4. Record your data in a table.	1. Plant three identical seeds in each of four cups containing the same amount and type of soil. 2. Add 50 milliliters (mL) of water to each cup. 3. After the seeds sprout, examine the sprouts carefully with a magnifying lens. 4. Record a drawing of each of the sprouts in a chart.

Which statement **best** describes why activity 1 is an experiment and activity 2 is an investigation?

- Ⓐ Only activity 1 has a control and variables.
- Ⓑ Only activity 1 requires the adding of fertilizer.
- Ⓒ Activity 1 requires more trials than activity 2 does.
- Ⓓ Activity 1 takes more time to set up than activity 2 does.

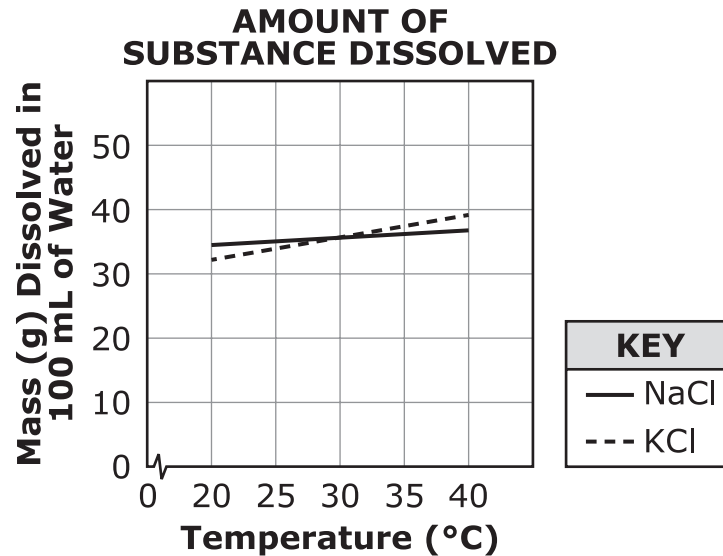
19. At a mid-ocean ridge, plates are pulled apart. Fracturing in the sea floor allows magma to move up from the mantle. The diagram below shows two plates that are moving apart



Which process in the rock cycle does this diagram **best** represent?

- Ⓐ chemical weathering of rocks
- Ⓑ erosion of sedimentary rocks
- Ⓒ compaction of melted rock
- Ⓓ formation of igneous rock

20. Students compared the solubility of sodium chloride (NaCl) and potassium chloride (KCl) in water at different temperatures. They recorded their results as mass, in grams (g), dissolved in 100 milliliters (mL) of water at different temperatures, in degrees Celsius ($^{\circ}\text{C}$), as shown in the graph below.



Which statement **best** compares the solubility of these substances?

- Ⓐ KCl is more soluble at 20°C than NaCl is.
- Ⓑ Both substances are equally soluble at 30°C .
- Ⓒ KCl will become less soluble at temperatures greater than 40°C .
- Ⓓ Both substances have the same solubility from 20°C through 40°C .

21. Alligators are found throughout Florida. Some of the characteristics of alligators are listed below.

- They are usually found in slow-flowing rivers.
- Females live in a small area.
- They eat fish, snails, birds, frogs, and mammals.
- Males live in areas of more than five square kilometers.
- They reduce their level of activity during winter.
- Females make large nests and protect their young.

Which limiting factor in an ecosystem would have the **greatest** impact on the alligator populations in Florida?

- Ⓐ reduction of shelter and nesting sites
- Ⓑ diseases that affect a certain frog species
- Ⓒ competition among male alligators for a mate
- Ⓓ increase in the rate of water flowing in a river

22. Wanda lists some heat transfers that can occur in an ocean, as shown below.

1. Warm water moves upward, and cold water moves downward.
2. Energy from the Sun heats the ocean's surface.
3. Warm air transfers heat to the ocean as it blows across the ocean's surface.
4. A stream of hot water from an opening on the ocean floor heats the water that touches the stream.

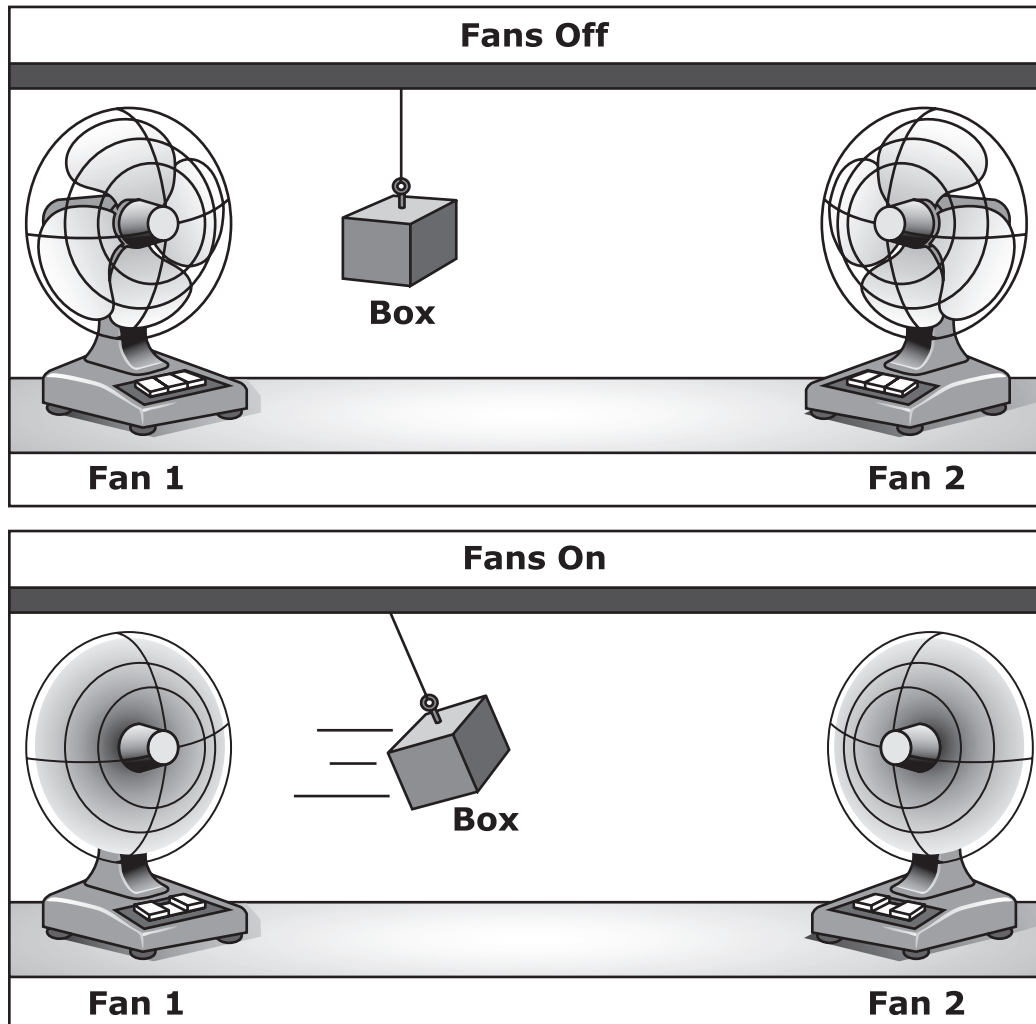
Which description from Wanda's list **best** represents radiation?

- Ⓐ 1
- Ⓑ 2
- Ⓒ 3
- Ⓓ 4

23. Scientists who investigate microscopic fossils are known as micropaleontologists. Which of the following would these scientists **most likely** study?

- Ⓐ the rocks that identify the location of ancient continents
- Ⓑ the evidence of changes in the climate of ancient Earth
- Ⓒ the different elevations of the surface of ancient Earth
- Ⓓ the rocks that contain evidence of ancient organisms

24. Joanna and Mason hang a small box from a string and place two identical electric fans near the box. When they turn on both fans to the same setting, Joanna and Mason observe that the box begins to swing toward the right, as shown in the illustration below.



Joanna claims that she can balance the forces on the box without turning off the fans. Which of the following would **most likely** produce the change in motion necessary to support Joanna's claim?

- Ⓐ Change the angle of Fan 2 so that it is blowing air onto the bottom of the box.
- Ⓑ Move Fan 2 so that both fans are blowing air onto the left side of the box.
- Ⓒ Change the setting of Fan 2 so that it is blowing less air than Fan 1 is.
- Ⓓ Move Fan 2 so that it is the same distance from the box as Fan 1 is.

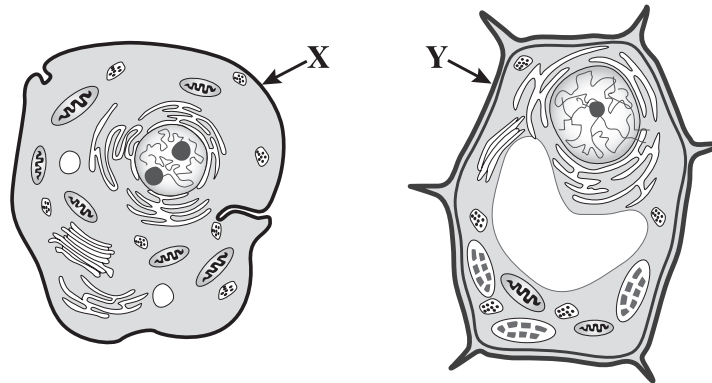
- 25.** The Andromeda Galaxy was previously called the Great Andromeda Nebula because astronomers thought this patch of light was composed of glowing gases. In the 1900s, scientists determined the nebula was actually a spiral galaxy. Which of the following describes the relationship between this galaxy and other astronomical bodies?
- Ⓐ This galaxy is the largest star and is found in the universe.
 - Ⓑ This galaxy is surrounded by thousands of stars in the universe.
 - Ⓒ This galaxy includes a universe that is composed of millions of stars.
 - Ⓓ This galaxy is one of billions of galaxies in the universe and contains billions of stars.

- 26.** Throughout history, scientific knowledge has changed. Which of the following resulted in an important change in scientific knowledge?
- Ⓐ the observation of phase changes of the Moon
 - Ⓑ the discovery of the function of the DNA molecule
 - Ⓒ the development of a hypothesis describing possible life on Mars
 - Ⓓ the prediction of when a comet would be most clearly visible from Earth

27. Cassandra flashes a beam of white light toward a prism. The prism separates the white light into different colors. Which of the following causes the separation of the white light?

- Ⓐ The white light is concentrated.
- Ⓑ The white light is absorbed.
- Ⓒ The white light is refracted.
- Ⓓ The white light is reflected.

28. Nyles used a microscope to observe two different types of cells. Drawings of the two cells are shown below.



Which of the following **best** contrasts the functions of the cell parts labeled X and Y?

- Ⓐ Part X stores water for future use, whereas part Y releases water out of the cell.
- Ⓑ Part X holds the structures in place in the cell, whereas part Y keeps all molecules from leaving the cell.
- Ⓒ Part X provides energy for the cell, whereas part Y helps with photosynthesis in the presence of sunlight.
- Ⓓ Part X regulates substances that enter or exit the cell, whereas part Y provides protection and support to the cell.

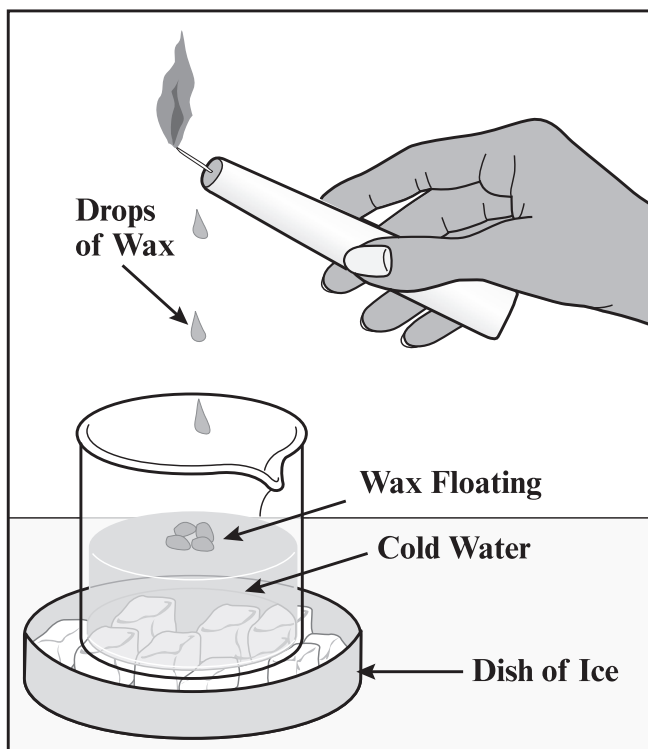
29. A student claims that studying rock layers on the side of a mountain would show that Earth has changed over geologic time. Which type of evidence from the rock layers **best** supports this claim?

- Ⓐ Some rock layers are solid, and other rock layers have holes that store water.
- Ⓑ Rock layers that are composed of sand and silt are found in different locations.
- Ⓒ Rock layers that are thick and smooth are deeper than rock layers that are thin and rough.
- Ⓓ Some rock layers contain fossils of plants, and other rock layers contain fossils of marine animals.

30. In 1980, a scientific journal published an article describing the observations that led to a hypothesis. The hypothesis stated that an asteroid striking Earth brought about the extinction of the dinosaurs. Which of the following would **best** provide support for a scientific hypothesis?

- Ⓐ opinions and personal beliefs
- Ⓑ new evidence and personal opinions
- Ⓒ new evidence and additional observations
- Ⓓ additional observations and informed predictions

31. Ms. Brooks demonstrated phase changes to her class. She lit a candle and let the melting wax drop into a beaker of cold water, as shown below. The temperature of the water was four degrees Celsius (4°C).



Which statement **best** explains what happened to the wax as it came in contact with the cold water in the beaker?

- Ⓐ The wax changed from a liquid to a solid as it gave off heat to its surroundings.
- Ⓑ The wax changed from a liquid to a solid as it increased in temperature.
- Ⓒ The wax remained in the liquid phase as it decreased in temperature.
- Ⓓ The wax remained in the liquid phase as it absorbed heat.

32. The planet Saturn takes 10,747 Earth days to complete one orbit around the Sun. Why does Saturn take longer to complete one orbit around the Sun than Earth does?

- Ⓐ Saturn has stronger gravity.
- Ⓑ Saturn is farther from the Sun.
- Ⓒ Saturn has a thicker atmosphere.
- Ⓓ Saturn is the only planet with rings.

33. A group of students slowly poured 100 milliliters (mL) of vegetable oil, 100 mL of colored water, and 100 mL of corn syrup into a 500-mL beaker. The students observed the beaker after 5 minutes and recorded their observations in their science notebooks. Which of the following are the students **most likely** trying to compare?

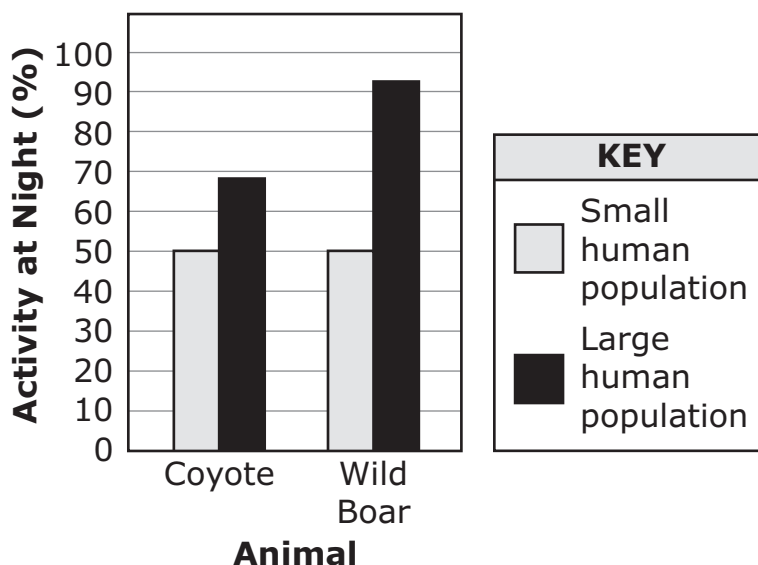
- Ⓐ the volume of the liquids
- Ⓑ the density of the liquids
- Ⓒ the weight of the liquids
- Ⓓ the color of the liquids

- 34.** A chemist observes a reaction between two substances in a closed container. Which of the following would **most likely** happen if the temperature is increased inside the container?
- Ⓐ The rate of the reaction would increase.
 - Ⓑ The mass of each of the atoms would increase.
 - Ⓒ The two substances would release nuclear energy.
 - Ⓓ The two substances would not change into different compounds.

Animal Activity

Coyotes and wild boars are animals that are active during the day and night. Both animals live in areas where human populations are increasing. Scientists collected data from various areas worldwide and concluded that the amount of coyote and wild boar activity is related to the population size of humans. The percentage of activity that occurs at night in different habitats is shown in Figure 1.

FIGURE 1: ANIMAL ACTIVITY IN TWO HABITATS



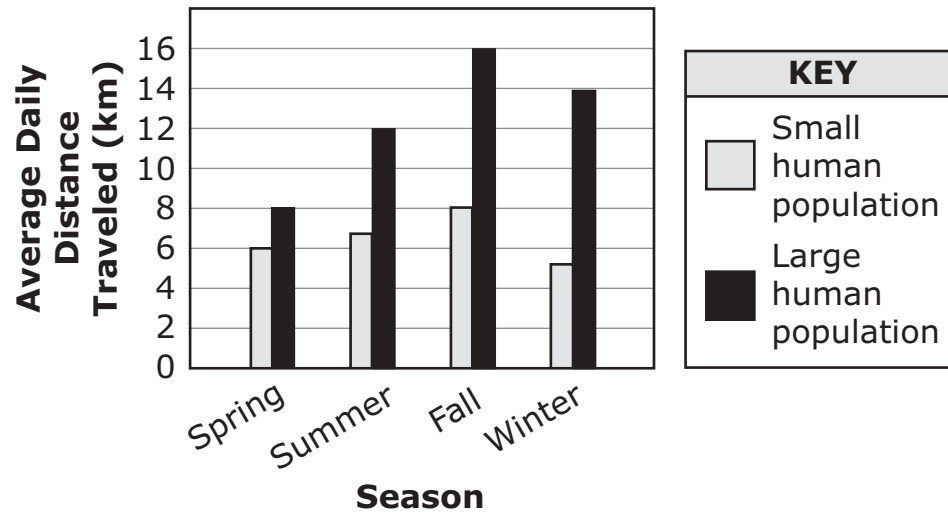
Scientists in California collected coyote droppings in habitats with small and large human populations. They used these data to determine the time of day and night that coyotes consumed prey. The scientists' data are shown in Table 1.

TABLE 1: PREY EATEN BY COYOTES IN TWO HABITATS

Human Population in Habitat	Percentage of Droppings That Contained Each Type of Prey	
	Prey Active during the Day	Prey Active during the Night
Small population	68%	81%
Large population	47%	100%

Scientists in Poland used radio collars to track the movements of wild boars in each type of habitat during different times of year. The average daily distances wild boars traveled in each habitat, in kilometers (km), are shown in Figure 2.

FIGURE 2: AVERAGE DAILY DISTANCE TRAVELED BY WILD BOARS



35. To answer this question, refer to the “Animal Activity” passage, figures, and table.

Which of these is the **most** important role of technology in the wild boar study?

- Ⓐ reducing bias during analysis and interpretation of the data
- Ⓑ producing graphs from the averages calculated by researchers
- Ⓒ collecting data over long periods of time and across great distances
- Ⓓ controlling external variables that are difficult for researchers to observe

36. To answer this question, refer to the “Animal Activity” passage, figures, and table.

Coyotes share their habitats with gray foxes, which are related to coyotes and are active at night. Gray foxes eat plants and animals, including some of the same animals that coyotes eat. Which of the following describes the **most likely** effect that coyotes have on gray foxes in locations with large human populations?

- Ⓐ decreased exposure to parasites and disease
- Ⓑ increased protection from larger predators
- Ⓒ decreased cooperation when finding food
- Ⓓ increased competition over prey species

37. To answer this question, refer to the “Animal Activity” passage, figures, and table.

Which statement describes how humans in an expanding city **most likely** affect the wild boars that live nearby?

- Ⓐ Humans decrease the effect that natural predators have on wild boars.
- Ⓑ Humans remove the plants that wild boars rely on for food and shelter.
- Ⓒ Humans create new ponds and lakes that wild boars must travel around.
- Ⓓ Humans increase the area of fields available for wild boars and other wildlife.

38. All plants and animals have threadlike structures in the nucleus of each cell. These structures contain a set of instructions in the form of DNA. Which of the following terms identifies these threadlike structures?

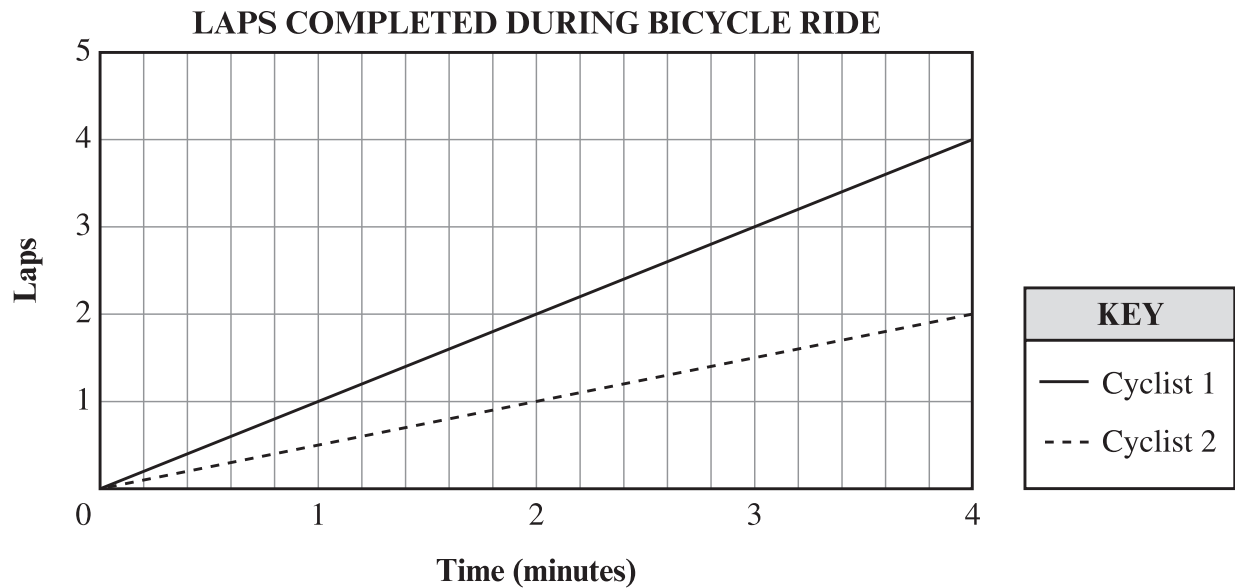
- Ⓐ alleles
- Ⓑ chromosomes
- Ⓒ cytoplasm
- Ⓓ mitochondria

39. Earth's atmosphere is important to the living organisms of the biosphere. Which of the following describes how the atmosphere **best** protects living organisms?

- Ⓐ by providing enough carbon dioxide in the ocean to support sea animals
- Ⓑ by changing oxygen into nitrogen, which can be used as a nutrient for animals
- Ⓒ by refracting visible light, allowing certain animals to see and capture their prey
- Ⓓ by trapping electromagnetic radiation, keeping temperatures within a range best suited for animal life

- 40.** A yeast is a single-celled organism that can reproduce sexually or asexually. Which of the following would confirm that a yeast had reproduced asexually?
- Ⓐ offspring with its own unique genetic information unlike that of the parent
 - Ⓑ offspring with the same genetic information as each of its two parents
 - Ⓒ offspring with half the genetic information from each of two parents
 - Ⓓ offspring with the same genetic information as the parent

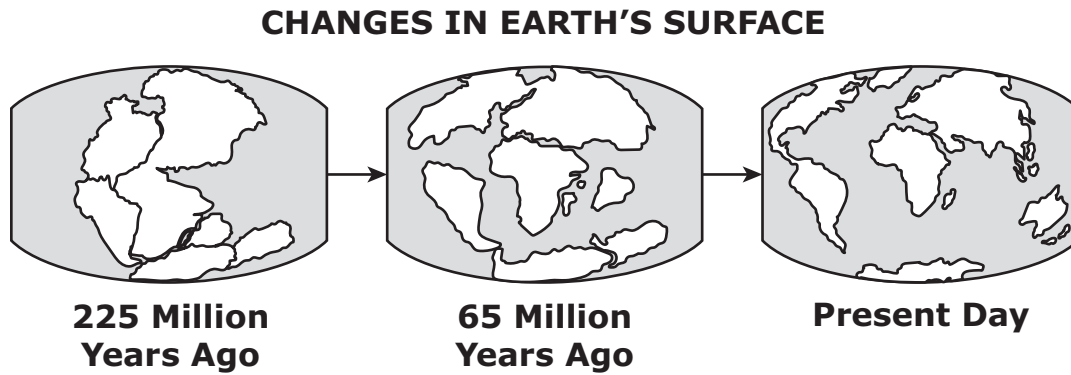
41. Two Olympic cyclists rode their bicycles on an oval track. The graph below represents the number of 500-meter laps they completed.



Based on the graph, which statement **best** describes the motion of the cyclists as they rode their bicycles?

- Ⓐ Both cyclists made stops about halfway through their rides.
- Ⓑ Both cyclists rode their bicycles at about the same speed.
- Ⓒ Cyclist 1 rode about twice as fast as Cyclist 2.
- Ⓓ Cyclist 2 made more stops than Cyclist 1.

42. Scientists have collected large amounts of evidence that support the idea shown in the illustration below.



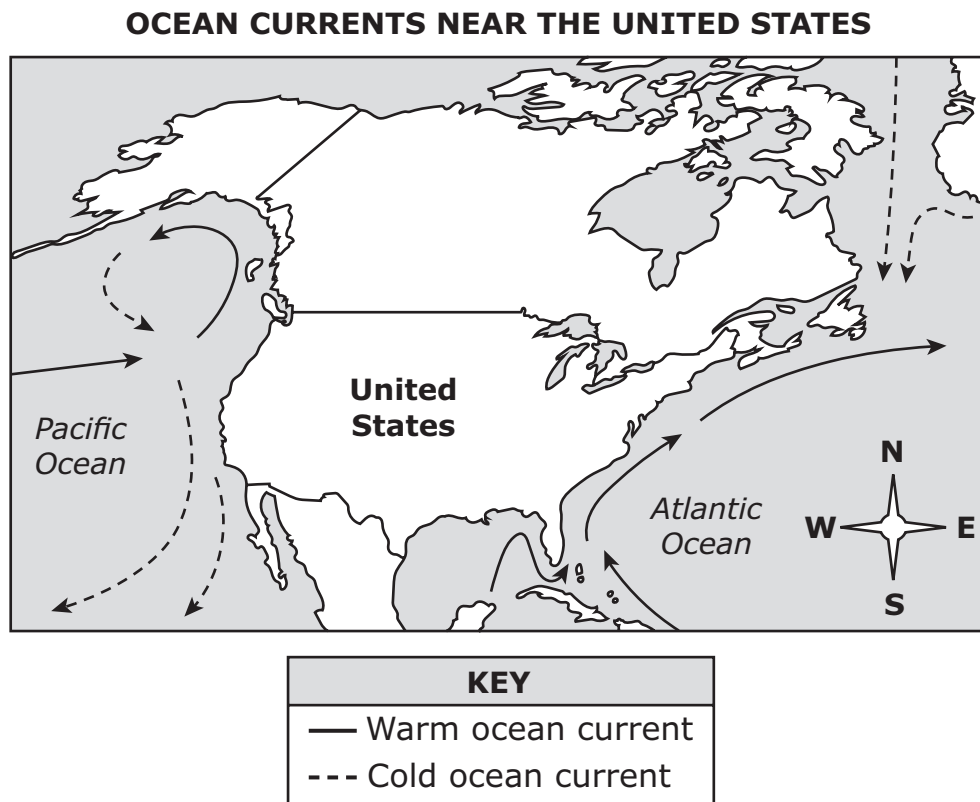
Which of the following is a scientific theory that is **best** supported by this evidence?

- Ⓐ evolution
- Ⓑ superposition
- Ⓒ plate tectonics
- Ⓓ universal gravitation

43. A scientist observes six lady beetles of the same species that live in the same location. Each of the lady beetles is a different shade of orange or yellow, and each has a different number of black spots. Which statement **most likely** explains how these differences relate to the evolution of the lady beetle species?

- Ⓐ These differences show that the species may learn to adapt to a warmer and drier climate.
- Ⓑ These differences show that damage to plants in the habitat has caused the species to develop additional spots.
- Ⓒ These differences show that mutations have developed in the population as the species was exposed to pollutants.
- Ⓓ These differences show that there is genetic variation in the population that may allow the species to survive changes to its habitat.

44. Ocean currents in the Pacific Ocean and the Atlantic Ocean affect climate and weather patterns in the United States. The map below shows some of these ocean currents.



A student researches climate data and determines that the eastern United States usually has higher humidity than the western United States does. Based on the ocean currents shown, which statement **best** explains the difference in humidity?

- Ⓐ The air mass in the Atlantic Ocean is warmer than the air mass in the Pacific Ocean is, and a warmer air mass holds more water vapor than a colder air mass does.
- Ⓑ The air mass in the Pacific Ocean is colder than the air mass in the Atlantic Ocean is, and a colder air mass causes more evaporation than a warmer air mass causes.
- Ⓒ The cold ocean current in the Pacific Ocean causes stronger winds than the warm ocean current in the Atlantic Ocean causes, and moisture evaporates from the Pacific Ocean air mass.
- Ⓓ The warm ocean current in the Atlantic Ocean produces denser air masses than the cold ocean current in the Pacific Ocean produces, and the dense air mass in the Atlantic Ocean cannot release precipitation.

- 45.** Dionne observes her science teacher testing a sample of an unknown substance. She recorded her observations in the table shown below.

Test	Characteristic Observed	Yes	No
1	Melts when heated		X
2	Breaks into smaller pieces	X	
3	Dissolves in water		X
4	Reacts with vinegar	X	

Which test would provide evidence of a chemical change?

- Ⓐ 1
- Ⓑ 2
- Ⓒ 3
- Ⓓ 4



Grade 8 Statewide Science Assessment

Test Release: 2024

1. Janice is learning about compounds. Which of the following **best** describes a compound?
- Ⓐ a substance made up purely of one element
 - Ⓑ a group of substances that are either acids or bases
 - Ⓒ a group of substances that can be separated by a physical change
 - Ⓓ a substance made up of two or more elements that are chemically combined

Answer Key: D

Percentage of Students Answering Correctly: 72%

Reporting Category: Physical Science

Benchmark: SC.8.P.8.8

Benchmark Description: Identify basic examples of and compare and classify the properties of compounds, including acids, bases, and salts.

2. Scientific theories are developed over time in all areas of science. Which statement **best** describes scientific theories?
- Ⓐ They are based on scientific laws that came first.
 - Ⓑ They are pieces of scientific evidence that support a hypothesis.
 - Ⓒ They are explanations of scientific events that may change based on continuous research.
 - Ⓓ They are predictions of events that usually remain constant because scientific ideas rarely change.

Answer Key: C

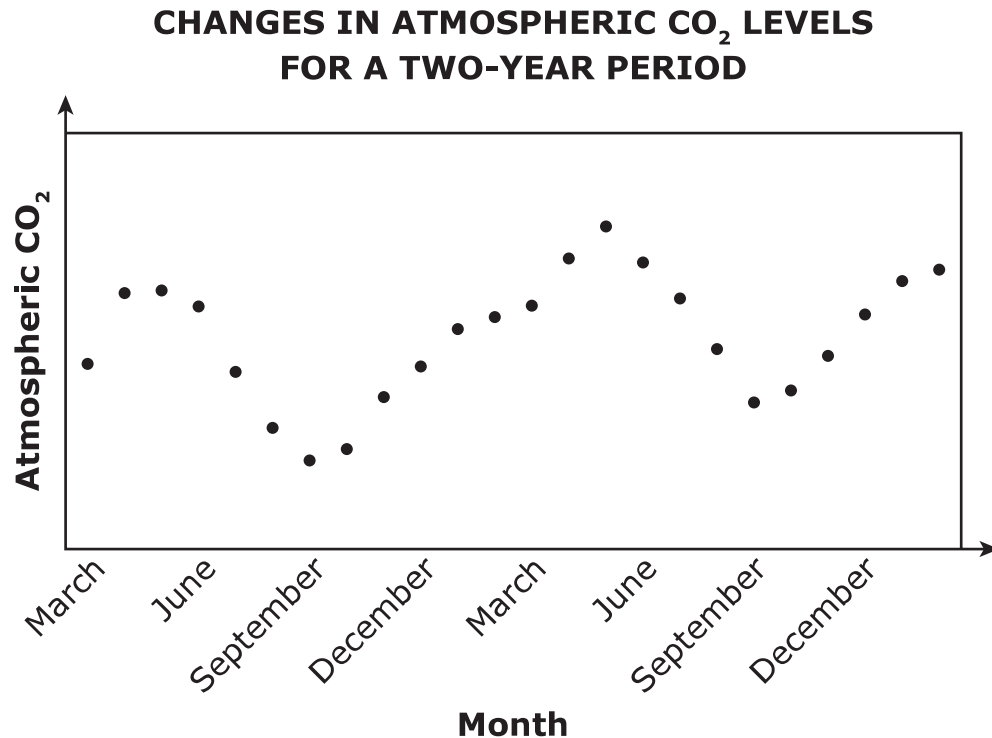
Percentage of Students Answering Correctly: 59%

Reporting Category: The Nature of Science

Benchmark: SC.8.N.3.2

Benchmark Description: Explain why theories may be modified but are rarely discarded.

3. Many interactions occur between the biosphere and the atmosphere. One of these interactions results in changes in atmospheric carbon dioxide (CO_2) levels. The graph below shows these changes over a two-year period.



Which of the following changes in the biosphere **most likely** caused these changes in the atmospheric CO_2 levels?

- Ⓐ Streams froze in the winter and melted in the spring.
- Ⓑ Plants grew leaves in the spring and lost leaves in the fall.
- Ⓒ Offspring were born in the summer and matured in the fall.
- Ⓓ Soil dried out in the summer and absorbed moisture in the winter.

Answer Key: B

Percentage of Students Answering Correctly: 47%

Reporting Category: Earth and Space Science

Benchmark: SC.6.E.7.4

Benchmark Description: Differentiate and show interactions among the geosphere, hydrosphere, cryosphere, atmosphere, and biosphere.

4. Jordan adds some soil and plants to a large glass jar to build a terrarium. He waters the plants and then adds two snails, two crickets, and an earthworm to the jar. Jordan then places the lid tightly on the jar and sets it on a sunny windowsill. Based on the law of conservation of mass, which of the following is expected to occur inside Jordan's terrarium?
- Ⓐ The mass will increase as the plants grow and decrease when the snails eat the plants.
 - Ⓑ The mass will increase as the earthworm digests the soil.
 - Ⓒ The mass will decrease when the crickets die.
 - Ⓓ The mass will neither increase nor decrease.

Answer Key: D

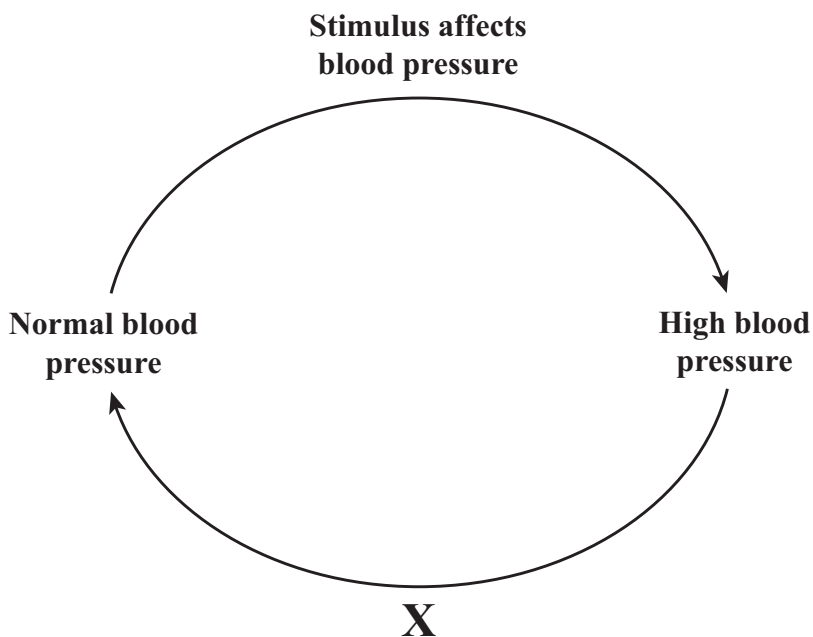
Percentage of Students Answering Correctly: 40%

Reporting Category: Life Science

Benchmark: SC.8.L.18.4

Benchmark Description: Cite evidence that living systems follow the Laws of Conservation of Mass and Energy.

5. Human body systems interact to keep blood pressure within a normal range. The diagram below represents this interaction.



Which of the following **best** describes what is occurring in part X of the diagram?

- Ⓐ Respiration remains high as other body systems work with the circulatory system.
- Ⓑ Blood pressure is regulated as the circulatory system shuts down other body systems.
- Ⓒ Blood pressure remains high as the circulatory system shuts down other body systems.
- Ⓓ Homeostasis is maintained as the other body systems work with the circulatory system.

Florida Test Release Support Document

Answer Key: D

Percentage of Students Answering Correctly: 37%

Reporting Category: Life Science

Benchmark: SC.6.L.14.5

Benchmark Description: Identify and investigate the general functions of the major systems of the human body (digestive, respiratory, circulatory, reproductive, excretory, immune, nervous, and musculoskeletal) and describe ways these systems interact with each other to maintain homeostasis.

6. The table below shows the range of frequencies, in Hertz (Hz), for part of the electromagnetic spectrum.

Region	Frequency (Hz)
Infrared	$3.0 \times 10^{12} - 4.3 \times 10^{14}$
Ultraviolet	$7.5 \times 10^{14} - 3.0 \times 10^{17}$
X-rays	$3.0 \times 10^{17} - 3.0 \times 10^{19}$
Gamma rays	$> 3.0 \times 10^{19}$

Which of the following types of radiation has the longest wavelength?

- Ⓐ Infrared
- Ⓑ Ultraviolet
- Ⓒ X-rays
- Ⓓ Gamma rays

Answer Key: A

Percentage of Students Answering Correctly: 25%

Reporting Category: Physical Science (Earth and Space Science)

Benchmark: SC.8.E.5.11

Benchmark Description: Identify and compare characteristics of the electromagnetic spectrum such as wavelength, frequency, use, and hazards and recognize its application to an understanding of planetary images and satellite photographs.

7. An astronomer observes a sunspot. Which of the following describes this feature of the Sun?
- Ⓐ a location where heat rises to the Sun's surface
 - Ⓑ a location that produces most of the Sun's heat energy
 - Ⓒ a location that is cooler than other parts of the Sun's surface
 - Ⓓ a location where energy and matter leave the Sun's atmosphere

Answer Key: C

Percentage of Students Answering Correctly: 36%

Reporting Category: Earth and Space Science

Benchmark: SC.8.E.5.6

Benchmark Description: Create models of solar properties, including rotation, structure of the Sun, convection, sunspots, solar flares, and prominences.

8. Ricky wants to finish constructing a chart that helps classify two organisms. The chart is shown below.

Question	Organism X	Organism Y
1. Does it require nutrients to survive?		
2. Does it have DNA?		
3. Do its cells lack cell walls?		
4. Can it reproduce?		

Accurately answering which question would **best** help distinguish these organisms as members of the animal kingdom rather than the plant or fungi kingdom?

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Answer Key: C

Percentage of Students Answering Correctly: 51%

Reporting Category: Life Science

Benchmark: SC.6.L.15.1

Benchmark Description: Analyze and describe how and why organisms are classified according to shared characteristics, with emphasis on the Linnaean system combined with the concept of Domains.

9. Atoms are made up of subatomic particles, including neutrons. Which of the following describes a neutron?
- Ⓐ has a greater volume than the electron cloud
 - Ⓑ has more mass than the nucleus
 - Ⓒ has no charge
 - Ⓓ has no mass

Answer Key: C

Percentage of Students Answering Correctly: 62%

Reporting Category: Physical Science

Benchmark: SC.8.P.8.7

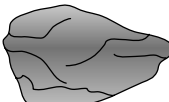
Benchmark Description: Explore the scientific theory of atoms (also known as atomic theory) by recognizing that atoms are the smallest unit of an element and are composed of sub-atomic particles (electrons surrounding a nucleus containing protons and neutrons).

10. Two groups of students hypothesized that rocks on Earth’s surface are damaged by acid rain.

- Group X tested the hypothesis by examining properties of some rocks and pouring a weak acid, vinegar, on each of the rocks.
- Group Y later repeated the same procedure with a different group of rocks.

The rocks used for the investigations and the results from each student group are shown below.

ROCKS TESTED

Group	Rock 1	Rock 2	Rock 3	Rock 4
X	 Calcite	 Dolostone	 Shale	 Limestone
Y	 Calcite	 Dolostone	 Dolostone	 Calcite

OBSERVATIONS OF THE ROCKS

Group	Number of Rocks Used	Average Mass of Rocks Used (grams)	Texture of Rocks	Reaction to Vinegar
X	4	46	3 rocks smooth 1 rock rough	3 rocks bubbled
Y	4	42	2 rocks smooth 2 rocks rough	4 rocks bubbled

Which of the following **best** describes an advantage Group X has over Group Y in supporting the hypothesis?

- Ⓐ Group X tested the rocks before Group Y did.
- Ⓑ Group X tested more rocks that are smoother.
- Ⓒ Group X tested rocks with a greater average mass.
- Ⓓ Group X tested a greater variety of rocks than Group Y did.

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Answer Key: D

Percentage of Students Answering Correctly: 50%

Reporting Category: The Nature of Science

Benchmark: SC.6.N.1.4

Benchmark Description: Discuss, compare, and negotiate methods used, results obtained, and explanations among groups of students conducting the same investigation.

11. Mandy applies the brakes on her bicycle and stops quickly to avoid hitting a basketball that bounces into the street. Which of the following **best** explains what happens to the bicycle's kinetic energy?

- Ⓐ It increases because the tires make a screeching noise.
- Ⓑ It is destroyed because Mandy stops the bicycle quickly.
- Ⓒ It remains the same because Mandy is the source of energy.
- Ⓓ It is transformed to heat because the tires are slowed by friction.

Answer Key: D

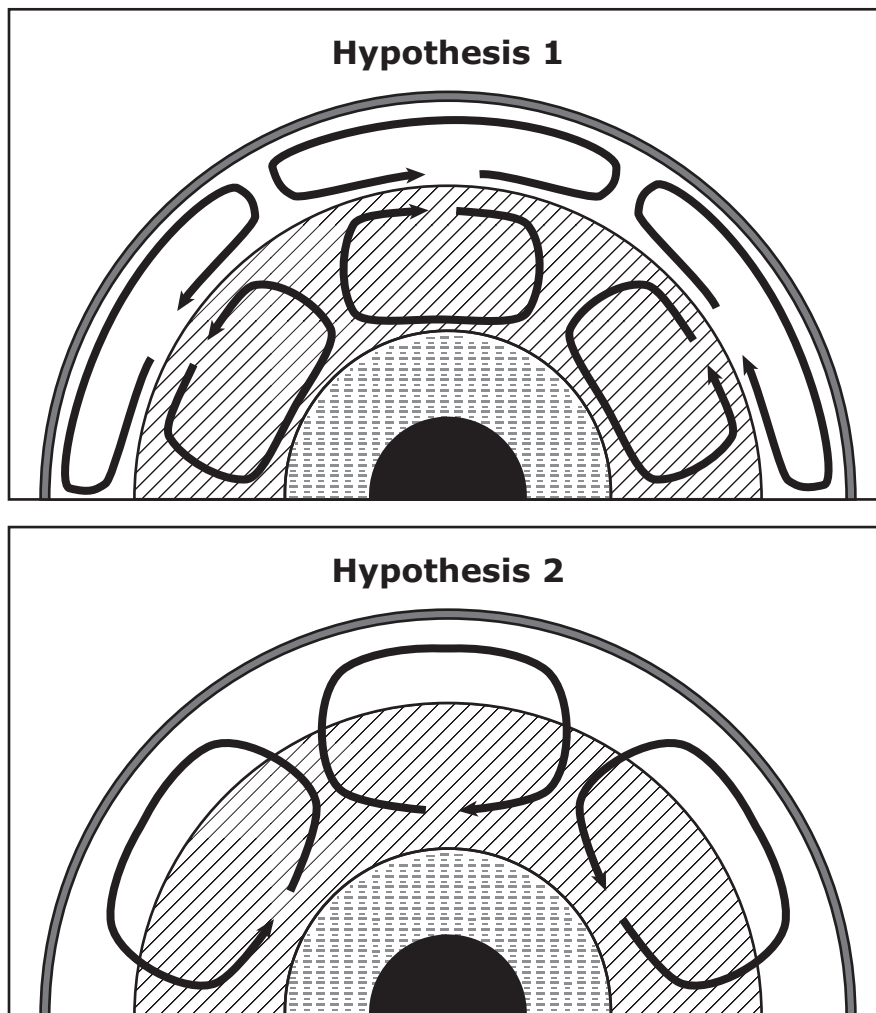
Percentage of Students Answering Correctly: 59%

Reporting Category: Physical Science

Benchmark: SC.7.P.11.3

Benchmark Description: Cite evidence to explain that energy cannot be created nor destroyed, only changed from one form to another.

12. The diagram shows two hypotheses for the same process that occurs beneath Earth's surface.



Which of the following processes is represented by these hypotheses?

- Ⓐ convection current movement in Earth's mantle
- Ⓑ plate movements in Earth's outer core
- Ⓒ layer separation in Earth's inner core
- Ⓓ rock formation in Earth's lithosphere

Answer Key: A

Percentage of Students Answering Correctly: 57%

Reporting Category: Earth and Space Science

Benchmark: SC.7.E.6.1

Benchmark Description: Describe the layers of the solid Earth, including the lithosphere, the hot convecting mantle, and the dense metallic liquid and solid cores.

13. North Atlantic right whales live in the ocean near the east coast of the United States and Canada. Fewer than 500 of these whales exist. Which of the following would contribute to the extinction of the North Atlantic right whales?

- Ⓐ Species that compete with right whales move to new habitats.
- Ⓑ Humans develop breeding programs for right whales in aquariums.
- Ⓒ The environment changes faster than right whale populations can adapt.
- Ⓓ New technologies prevent right whales from being caught in fishing gear.

Answer Key: C

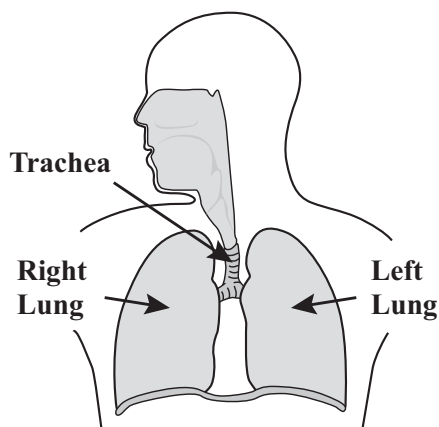
Percentage of Students Answering Correctly: 71%

Reporting Category: Life Science

Benchmark: SC.7.L.15.3

Benchmark Description: Explore the scientific theory of evolution by relating how the inability of a species to adapt within a changing environment may contribute to the extinction of that species.

14. The diagram below identifies three structures found in the human body.



Which level of organization describes each of these three structures?

- Ⓐ cell
- Ⓑ organ
- Ⓒ organism
- Ⓓ tissue

Answer Key: B

Percentage of Students Answering Correctly: 73%

Reporting Category: Life Science

Benchmark: SC.6.L.14.1

Benchmark Description: Describe and identify patterns in the hierarchical organization of organisms from atoms to molecules and cells to tissues to organs to organ systems to organisms.

15. A space probe lands on a planet that has half the pull of gravity as Earth. What effect does this have on the space probe's weight and mass?

- ☒ Ⓐ Its weight is reduced to half its weight on Earth, and its mass remains the same as on Earth.
- ☐ Ⓑ Its weight remains the same as on Earth, and its mass is reduced to half its mass on Earth.
- ☐ Ⓒ Both its weight and mass are half of what they are on Earth.
- ☐ Ⓓ Both its weight and mass remain the same as on Earth.

Answer Key: A

Percentage of Students Answering Correctly: 63%

Reporting Category: Physical Science

Benchmark: SC.8.P.8.2

Benchmark Description: Differentiate between weight and mass recognizing that weight is the amount of gravitational pull on an object and is distinct from, though proportional to, mass.

16. Cells experience similar processes. One of these processes involves controlling cellular wastes. Why do cells undergo processes that control cellular wastes?

- Ⓐ All cells must release cellular wastes to maintain homeostasis.
- Ⓑ Plant cells eliminate cellular wastes to regulate the rate of cell reproduction.
- Ⓒ Animal cells extract energy from cellular wastes to perform cellular functions.
- Ⓓ All cells must release cellular wastes to provide a means of maintaining cell size.

Answer Key: A

Percentage of Students Answering Correctly: 55%

Reporting Category: Life Science

Benchmark: SC.6.L.14.3

Benchmark Description: Recognize and explore how cells of all organisms undergo similar processes to maintain homeostasis, including extracting energy from food, getting rid of waste, and reproducing.

17. Nancy observes the Moon from the same location outside each night for one week. She notices that when it first appears each night, the Moon is a little farther east than it was the night before. Which of the following is one explanation for her observation?

- Ⓐ The Moon's speed of rotation changes as it revolves around Earth.
- Ⓑ The Moon revolves around Earth as Earth revolves around the Sun.
- Ⓒ The Moon and Earth take the same amount of time to complete one rotation.
- Ⓓ The Moon completes one rotation each time Earth completes one revolution.

Answer Key: B

Percentage of Students Answering Correctly: 61%

Reporting Category: Earth and Space Science

Benchmark: SC.8.E.5.9

Benchmark Description: Explain the impact of objects in space on each other including:
1. the Sun on the Earth including seasons and gravitational attraction
2. the Moon on the Earth, including phases, tides, and eclipses, and the relative position of each body.

- 18.** Brittany is investigating plant growth. She completes the following activities in class.

Activity 1	Activity 2
1. Plant three identical seeds in each of four cups containing the same amount and type of soil. 2. Add different amounts of the same fertilizer to three of the cups and 50 milliliters (mL) of water to all of the cups. 3. When the seeds sprout, measure the height of the plants in each cup. 4. Record your data in a table.	1. Plant three identical seeds in each of four cups containing the same amount and type of soil. 2. Add 50 milliliters (mL) of water to each cup. 3. After the seeds sprout, examine the sprouts carefully with a magnifying lens. 4. Record a drawing of each of the sprouts in a chart.

Which statement **best** describes why activity 1 is an experiment and activity 2 is an investigation?

- ☒ (A) Only activity 1 has a control and variables.
- ☐ (B) Only activity 1 requires the adding of fertilizer.
- ☐ (C) Activity 1 requires more trials than activity 2 does.
- ☐ (D) Activity 1 takes more time to set up than activity 2 does.

Answer Key: A

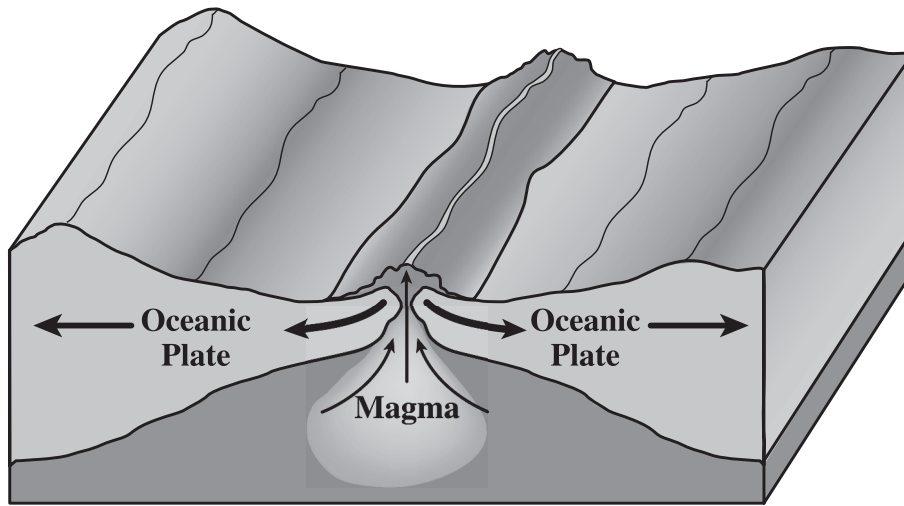
Percentage of Students Answering Correctly: 53%

Reporting Category: The Nature of Science

Benchmark: SC.7.N.1.3

Benchmark Description: Distinguish between an experiment (which must involve the identification and control of variables) and other forms of scientific investigation, and explain that not all scientific knowledge is derived from experimentation.

19. At a mid-ocean ridge, plates are pulled apart. Fracturing in the sea floor allows magma to move up from the mantle. The diagram below shows two plates that are moving apart



Which process in the rock cycle does this diagram **best** represent?

- Ⓐ chemical weathering of rocks
- Ⓑ erosion of sedimentary rocks
- Ⓒ compaction of melted rock
- Ⓓ formation of igneous rock

Answer Key: D

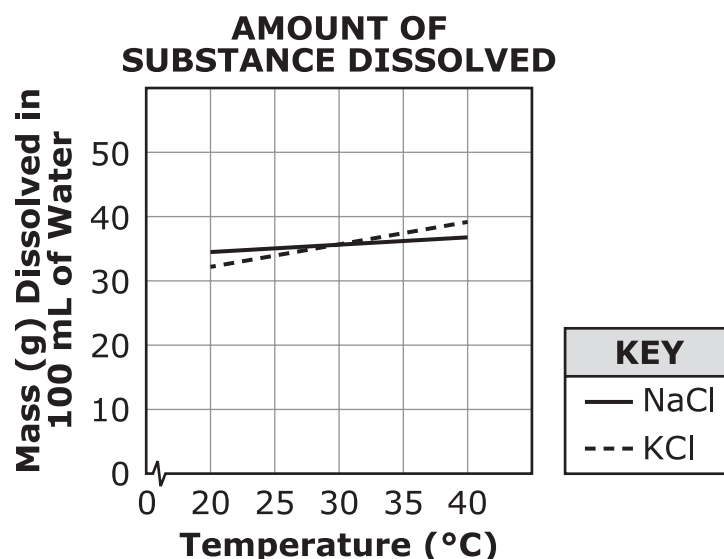
Percentage of Students Answering Correctly: 52%

Reporting Category: Earth and Space Science

Benchmark: SC.7.E.6.2

Benchmark Description: Identify the patterns within the rock cycle and relate them to surface events (weathering and erosion) and subsurface events (plate tectonics and mountain building).

- 20.** Students compared the solubility of sodium chloride (NaCl) and potassium chloride (KCl) in water at different temperatures. They recorded their results as mass, in grams (g), dissolved in 100 milliliters (mL) of water at different temperatures, in degrees Celsius ($^{\circ}\text{C}$), as shown in the graph below.



Which statement **best** compares the solubility of these substances?

- Ⓐ KCl is more soluble at 20°C than NaCl is.
- Ⓑ Both substances are equally soluble at 30°C .
- Ⓒ KCl will become less soluble at temperatures greater than 40°C .
- Ⓓ Both substances have the same solubility from 20°C through 40°C .

Answer Key: B

Percentage of Students Answering Correctly: 52%

Reporting Category: Physical Science

Benchmark: SC.8.P.8.4

Benchmark Description: Classify and compare substances on the basis of characteristic physical properties that can be demonstrated or measured; for example, density, thermal or electrical conductivity, solubility, magnetic properties, melting and boiling points, and know that these properties are independent of the amount of the sample.

21. Alligators are found throughout Florida. Some of the characteristics of alligators are listed below.

- They are usually found in slow-flowing rivers.
- Females live in a small area.
- They eat fish, snails, birds, frogs, and mammals.
- Males live in areas of more than five square kilometers.
- They reduce their level of activity during winter.
- Females make large nests and protect their young.

Which limiting factor in an ecosystem would have the **greatest** impact on the alligator populations in Florida?

- ☒ (A) reduction of shelter and nesting sites
- ☐ (B) diseases that affect a certain frog species
- ☐ (C) competition among male alligators for a mate
- ☐ (D) increase in the rate of water flowing in a river

Answer Key: A

Percentage of Students Answering Correctly: 56%

Reporting Category: Life Science

Benchmark: SC.7.L.17.3

Benchmark Description: Describe and investigate various limiting factors in the local ecosystem and their impact on native populations, including food, shelter, water, space, disease, parasitism, predation, and nesting sites.

22. Wanda lists some heat transfers that can occur in an ocean, as shown below.

1. Warm water moves upward, and cold water moves downward.
2. Energy from the Sun heats the ocean's surface.
3. Warm air transfers heat to the ocean as it blows across the ocean's surface.
4. A stream of hot water from an opening on the ocean floor heats the water that touches the stream.

Which description from Wanda's list **best** represents radiation?

- Ⓐ 1
- Ⓑ 2
- Ⓒ 3
- Ⓓ 4

Answer Key: B

Percentage of Students Answering Correctly: 66%

Reporting Category: Earth and Space Science

Benchmark: SC.6.E.7.1

Benchmark Description: Differentiate among radiation, conduction, and convection, the three mechanisms by which heat is transferred through Earth's system.

23. Scientists who investigate microscopic fossils are known as micropaleontologists. Which of the following would these scientists **most likely** study?

- Ⓐ the rocks that identify the location of ancient continents
- Ⓑ the evidence of changes in the climate of ancient Earth
- Ⓒ the different elevations of the surface of ancient Earth
- Ⓓ the rocks that contain evidence of ancient organisms

Answer Key: D

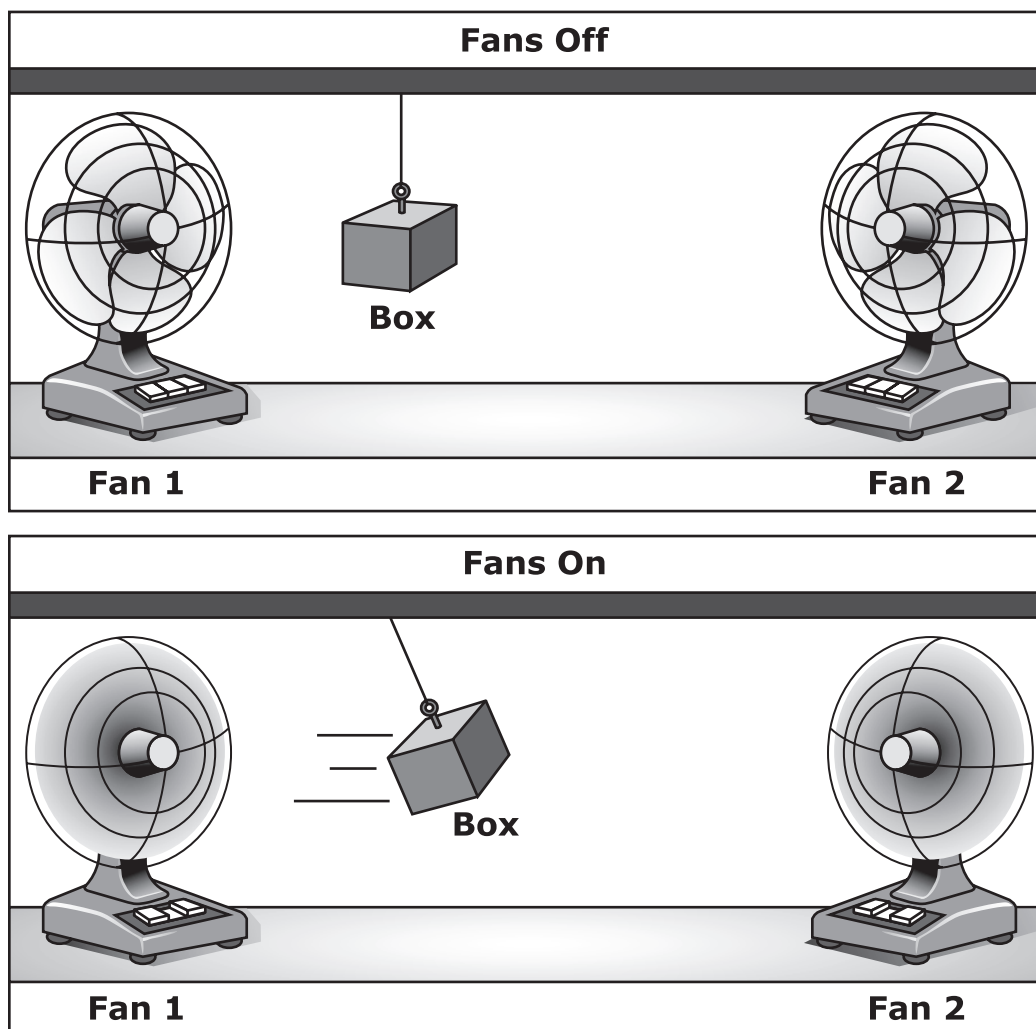
Percentage of Students Answering Correctly: 72%

Reporting Category: The Nature of Science

Benchmark: SC.7.N.1.5

Benchmark Description: Describe the methods used in the pursuit of a scientific explanation as seen in different fields of science such as biology, geology, and physics.

24. Joanna and Mason hang a small box from a string and place two identical electric fans near the box. When they turn on both fans to the same setting, Joanna and Mason observe that the box begins to swing toward the right, as shown in the illustration below.



Joanna claims that she can balance the forces on the box without turning off the fans. Which of the following would **most likely** produce the change in motion necessary to support Joanna's claim?

- Ⓐ Change the angle of Fan 2 so that it is blowing air onto the bottom of the box.
- Ⓑ Move Fan 2 so that both fans are blowing air onto the left side of the box.
- Ⓒ Change the setting of Fan 2 so that it is blowing less air than Fan 1 is.
- Ⓓ Move Fan 2 so that it is the same distance from the box as Fan 1 is.

Florida Test Release Support Document

Answer Key: D

Percentage of Students Answering Correctly: 77%

Reporting Category: Physical Science

Benchmark: SC.6.P.13.3

Benchmark Description: Investigate and describe that an unbalanced force acting on an object changes its speed, or direction of motion, or both.

25. The Andromeda Galaxy was previously called the Great Andromeda Nebula because astronomers thought this patch of light was composed of glowing gases. In the 1900s, scientists determined the nebula was actually a spiral galaxy. Which of the following describes the relationship between this galaxy and other astronomical bodies?

- Ⓐ This galaxy is the largest star and is found in the universe.
- Ⓑ This galaxy is surrounded by thousands of stars in the universe.
- Ⓒ This galaxy includes a universe that is composed of millions of stars.
- Ⓓ This galaxy is one of billions of galaxies in the universe and contains billions of stars.

Answer Key: D

Percentage of Students Answering Correctly: 68%

Reporting Category: Earth and Space Science

Benchmark: SC.8.E.5.2

Benchmark Description: Recognize that the universe contains many billions of galaxies and that each galaxy contains many billions of stars.

26. Throughout history, scientific knowledge has changed. Which of the following resulted in an important change in scientific knowledge?

- Ⓐ the observation of phase changes of the Moon
- Ⓑ the discovery of the function of the DNA molecule
- Ⓒ the development of a hypothesis describing possible life on Mars
- Ⓓ the prediction of when a comet would be most clearly visible from Earth

Answer Key: B

Percentage of Students Answering Correctly: 56%

Reporting Category: The Nature of Science

Benchmark: SC.7.N.2.1

Benchmark Description: Identify an instance from the history of science in which scientific knowledge has changed when new evidence or new interpretations are encountered.

27. Cassandra flashes a beam of white light toward a prism. The prism separates the white light into different colors. Which of the following causes the separation of the white light?

- Ⓐ The white light is concentrated.
- Ⓑ The white light is absorbed.
- Ⓒ The white light is refracted.
- Ⓓ The white light is reflected.

Answer Key: C

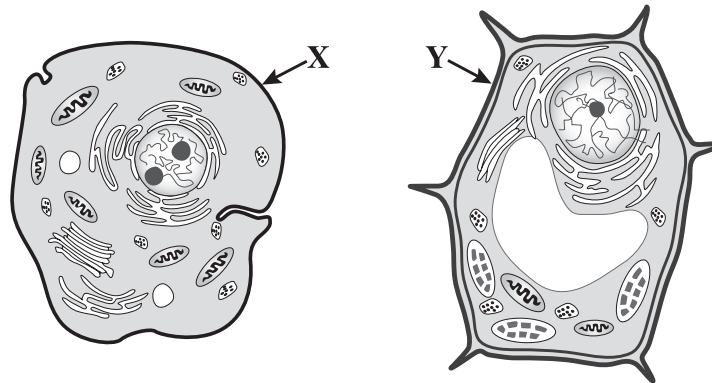
Percentage of Students Answering Correctly: 57%

Reporting Category: Physical Science

Benchmark: SC.7.P.10.2

Benchmark Description: Observe and explain that light can be reflected, refracted, and/or absorbed.

28. Nyles used a microscope to observe two different types of cells. Drawings of the two cells are shown below.



Which of the following **best** contrasts the functions of the cell parts labeled X and Y?

- Ⓐ Part X stores water for future use, whereas part Y releases water out of the cell.
- Ⓑ Part X holds the structures in place in the cell, whereas part Y keeps all molecules from leaving the cell.
- Ⓒ Part X provides energy for the cell, whereas part Y helps with photosynthesis in the presence of sunlight.
- Ⓓ Part X regulates substances that enter or exit the cell, whereas part Y provides protection and support to the cell.

Answer Key: D

Percentage of Students Answering Correctly: 56%

Reporting Category: Life Science

Benchmark: SC.6.L.14.4

Benchmark Description: Compare and contrast the structure and function of major organelles of plant and animal cells, including cell wall, cell membrane, nucleus, cytoplasm, chloroplasts, mitochondria, and vacuoles.

29. A student claims that studying rock layers on the side of a mountain would show that Earth has changed over geologic time. Which type of evidence from the rock layers **best** supports this claim?

- Ⓐ Some rock layers are solid, and other rock layers have holes that store water.
- Ⓑ Rock layers that are composed of sand and silt are found in different locations.
- Ⓒ Rock layers that are thick and smooth are deeper than rock layers that are thin and rough.
- Ⓓ Some rock layers contain fossils of plants, and other rock layers contain fossils of marine animals.

Answer Key: D

Percentage of Students Answering Correctly: 59%

Reporting Category: Earth and Space Science

Benchmark: SC.7.E.6.4

Benchmark Description: Explain and give examples of how physical evidence supports scientific theories that Earth has evolved over geologic time due to natural processes.

30. In 1980, a scientific journal published an article describing the observations that led to a hypothesis. The hypothesis stated that an asteroid striking Earth brought about the extinction of the dinosaurs. Which of the following would **best** provide support for a scientific hypothesis?

- Ⓐ opinions and personal beliefs
- Ⓑ new evidence and personal opinions
- Ⓒ new evidence and additional observations
- Ⓓ additional observations and informed predictions

Answer Key: C

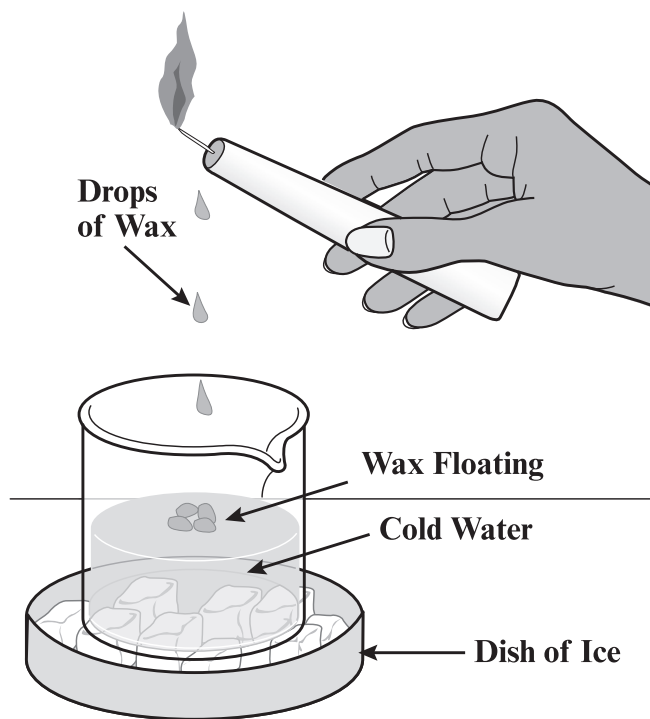
Percentage of Students Answering Correctly: 61%

Reporting Category: The Nature of Science

Benchmark: SC.8.N.1.6

Benchmark Description: Understand that scientific investigations involve the collection of relevant empirical evidence, the use of logical reasoning, and the application of imagination in devising hypotheses, predictions, explanations and models to make sense of the collected evidence.

31. Ms. Brooks demonstrated phase changes to her class. She lit a candle and let the melting wax drop into a beaker of cold water, as shown below. The temperature of the water was four degrees Celsius (4°C).



Which statement **best** explains what happened to the wax as it came in contact with the cold water in the beaker?

- Ⓐ The wax changed from a liquid to a solid as it gave off heat to its surroundings.
- Ⓑ The wax changed from a liquid to a solid as it increased in temperature.
- Ⓒ The wax remained in the liquid phase as it decreased in temperature.
- Ⓓ The wax remained in the liquid phase as it absorbed heat.

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Answer Key: A

Percentage of Students Answering Correctly: 68%

Reporting Category: Physical Science

Benchmark: SC.7.P.11.1

Benchmark Description: Recognize that adding heat to or removing heat from a system may result in a temperature change and possibly a change of state.

32. The planet Saturn takes 10,747 Earth days to complete one orbit around the Sun. Why does Saturn take longer to complete one orbit around the Sun than Earth does?

- Ⓐ Saturn has stronger gravity.
- Ⓑ Saturn is farther from the Sun.
- Ⓒ Saturn has a thicker atmosphere.
- Ⓓ Saturn is the only planet with rings.

Answer Key: B

Percentage of Students Answering Correctly: 76%

Reporting Category: Earth and Space Science

Benchmark: SC.8.E.5.7

Benchmark Description: Compare and contrast the properties of objects in the Solar System, including the Sun, planets, and moons to those of Earth, such as gravitational force, distance from the Sun, speed, movement, temperature, and atmospheric conditions.

33. A group of students slowly poured 100 milliliters (mL) of vegetable oil, 100 mL of colored water, and 100 mL of corn syrup into a 500-mL beaker. The students observed the beaker after 5 minutes and recorded their observations in their science notebooks. Which of the following are the students **most likely** trying to compare?

- Ⓐ the volume of the liquids
- Ⓑ the density of the liquids
- Ⓒ the weight of the liquids
- Ⓓ the color of the liquids

Answer Key: B

Percentage of Students Answering Correctly: 63%

Reporting Category: The Nature of Science

Benchmark: SC.8.N.1.1

Benchmark Description: Define a problem from the eighth grade curriculum using appropriate reference materials to support scientific understanding; plan and carry out scientific investigations of various types, such as systematic observations or experiments; identify variables; collect and organize data; interpret data in charts, tables, and graphics; analyze information; make predictions; and defend conclusions.

34. A chemist observes a reaction between two substances in a closed container. Which of the following would **most likely** happen if the temperature is increased inside the container?

- Ⓐ The rate of the reaction would increase.
- Ⓑ The mass of each of the atoms would increase.
- Ⓒ The two substances would release nuclear energy.
- Ⓓ The two substances would not change into different compounds.

Answer Key: A

Percentage of Students Answering Correctly: 65%

Reporting Category: Physical Science

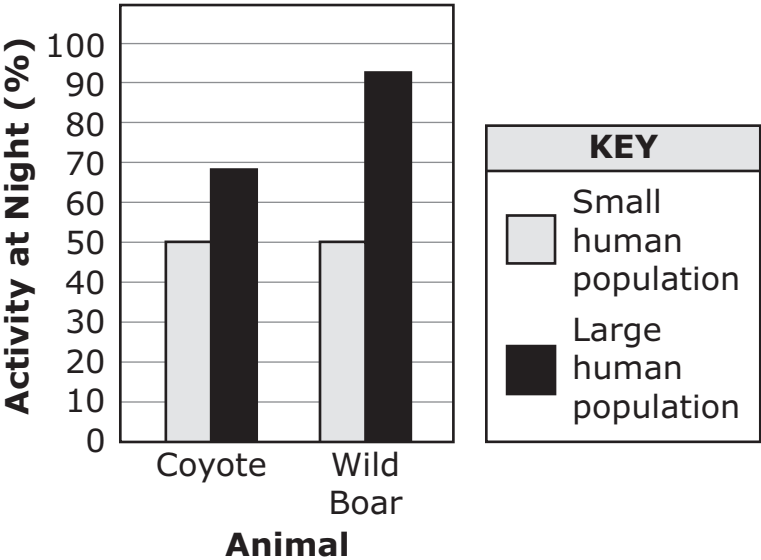
Benchmark: SC.8.P.9.3

Benchmark Description: Investigate and describe how temperature influences chemical changes.

Animal Activity

Coyotes and wild boars are animals that are active during the day and night. Both animals live in areas where human populations are increasing. Scientists collected data from various areas worldwide and concluded that the amount of coyote and wild boar activity is related to the population size of humans. The percentage of activity that occurs at night in different habitats is shown in Figure 1.

FIGURE 1: ANIMAL ACTIVITY IN TWO HABITATS



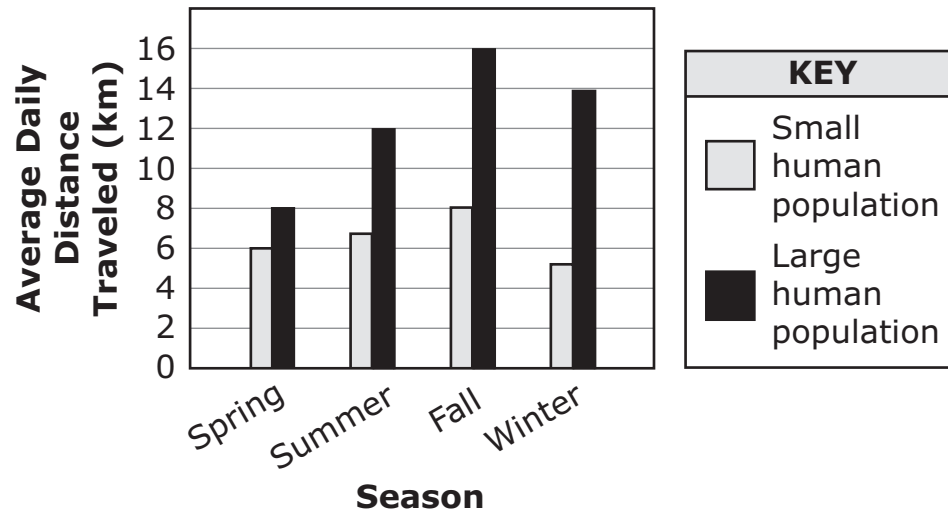
Scientists in California collected coyote droppings in habitats with small and large human populations. They used these data to determine the time of day and night that coyotes consumed prey. The scientists’ data are shown in Table 1.

TABLE 1: PREY EATEN BY COYOTES IN TWO HABITATS

Human Population in Habitat	Percentage of Droppings That Contained Each Type of Prey	
	Prey Active during the Day	Prey Active during the Night
Small population	68%	81%
Large population	47%	100%

Scientists in Poland used radio collars to track the movements of wild boars in each type of habitat during different times of year. The average daily distances wild boars traveled in each habitat, in kilometers (km), are shown in Figure 2.

FIGURE 2: AVERAGE DAILY DISTANCE TRAVELED BY WILD BOARS



35. To answer this question, refer to the “Animal Activity” passage, figures, and table.

Which of these is the **most** important role of technology in the wild boar study?

- Ⓐ reducing bias during analysis and interpretation of the data
- Ⓑ producing graphs from the averages calculated by researchers
- Ⓒ collecting data over long periods of time and across great distances
- Ⓓ controlling external variables that are difficult for researchers to observe

Answer Key: C

Percentage of Students Answering Correctly: 56%

Reporting Category: The Nature of Science (Earth and Space Science)

Benchmark: SC.8.E.5.10

Benchmark Description: Assess how technology is essential to science for such purposes as access to outer space and other remote locations, sample collection, measurement, data collection and storage, computation, and communication of information.

36. To answer this question, refer to the “Animal Activity” passage, figures, and table.

Coyotes share their habitats with gray foxes, which are related to coyotes and are active at night. Gray foxes eat plants and animals, including some of the same animals that coyotes eat. Which of the following describes the **most likely** effect that coyotes have on gray foxes in locations with large human populations?

- Ⓐ decreased exposure to parasites and disease
- Ⓑ increased protection from larger predators
- Ⓒ decreased cooperation when finding food
- Ⓓ increased competition over prey species

Answer Key: D

Percentage of Students Answering Correctly: 63%

Reporting Category: Life Science

Benchmark: SC.7.L.17.2

Benchmark Description: Compare and contrast the relationships among organisms, such as mutualism, predation, parasitism, competition, and commensalism.

37. To answer this question, refer to the “Animal Activity” passage, figures, and table.

Which statement describes how humans in an expanding city **most likely** affect the wild boars that live nearby?

- Ⓐ Humans decrease the effect that natural predators have on wild boars.
- Ⓑ Humans remove the plants that wild boars rely on for food and shelter.
- Ⓒ Humans create new ponds and lakes that wild boars must travel around.
- Ⓓ Humans increase the area of fields available for wild boars and other wildlife.

Answer Key: B

Percentage of Students Answering Correctly: 54%

Reporting Category: Earth and Space Science

Benchmark: SC.7.E.6.6

Benchmark Description: Identify the impact that humans have had on Earth, such as deforestation, urbanization, desertification, erosion, air and water quality, and changing the flow of water.

38. All plants and animals have threadlike structures in the nucleus of each cell. These structures contain a set of instructions in the form of DNA. Which of the following terms identifies these threadlike structures?

- Ⓐ alleles
- Ⓑ chromosomes
- Ⓒ cytoplasm
- Ⓓ mitochondria

Answer Key: B

Percentage of Students Answering Correctly: 51%

Reporting Category: Life Science

Benchmark: SC.7.L.16.1

Benchmark Description: Understand and explain that every organism requires a set of instructions that specifies its traits, that this hereditary information (DNA) contains genes located in the chromosomes of each cell, and that heredity is the passage of these instructions from one generation to another.

39. Earth's atmosphere is important to the living organisms of the biosphere. Which of the following describes how the atmosphere **best** protects living organisms?

- Ⓐ by providing enough carbon dioxide in the ocean to support sea animals
- Ⓑ by changing oxygen into nitrogen, which can be used as a nutrient for animals
- Ⓒ by refracting visible light, allowing certain animals to see and capture their prey
- Ⓓ by trapping electromagnetic radiation, keeping temperatures within a range best suited for animal life

Answer Key: D

Percentage of Students Answering Correctly: 68%

Reporting Category: Earth and Space Science

Benchmark: SC.6.E.7.9

Benchmark Description: Describe how the composition and structure of the atmosphere protects life and insulates the planet.

- 40.** A yeast is a single-celled organism that can reproduce sexually or asexually. Which of the following would confirm that a yeast had reproduced asexually?
- Ⓐ offspring with its own unique genetic information unlike that of the parent
 - Ⓑ offspring with the same genetic information as each of its two parents
 - Ⓒ offspring with half the genetic information from each of two parents
 - Ⓓ offspring with the same genetic information as the parent

Answer Key: D

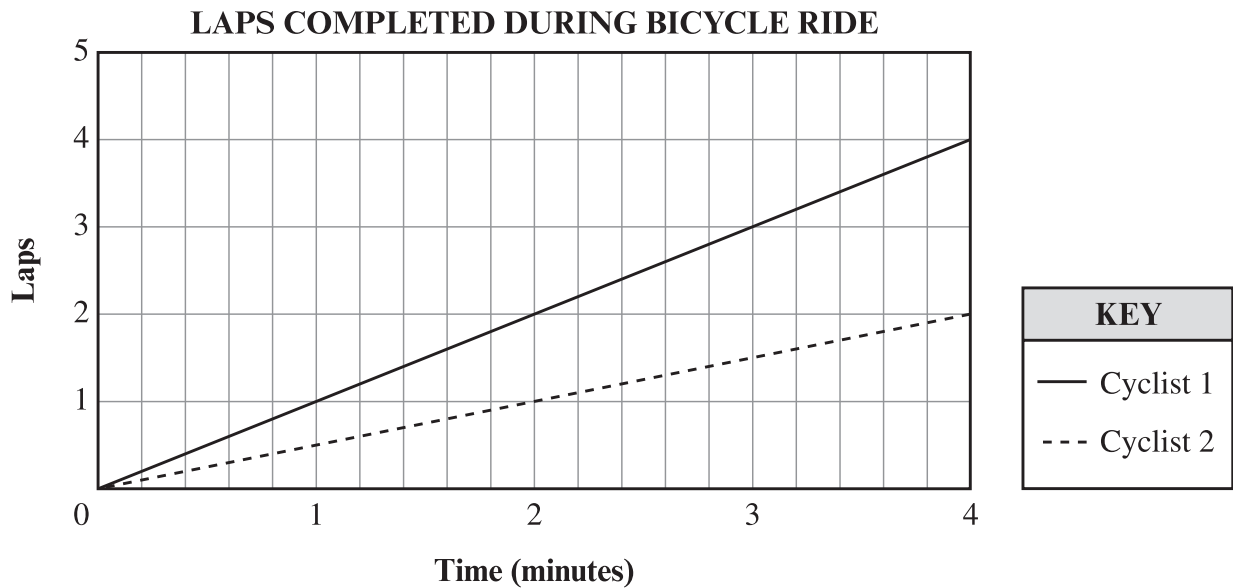
Percentage of Students Answering Correctly: 54%

Reporting Category: Life Science

Benchmark: SC.7.L.16.3

Benchmark Description: Compare and contrast the general processes of sexual reproduction requiring meiosis and asexual reproduction requiring mitosis.

41. Two Olympic cyclists rode their bicycles on an oval track. The graph below represents the number of 500-meter laps they completed.



Based on the graph, which statement **best** describes the motion of the cyclists as they rode their bicycles?

- Ⓐ Both cyclists made stops about halfway through their rides.
- Ⓑ Both cyclists rode their bicycles at about the same speed.
- Ⓒ Cyclist 1 rode about twice as fast as Cyclist 2.
- Ⓓ Cyclist 2 made more stops than Cyclist 1.

Answer Key: C

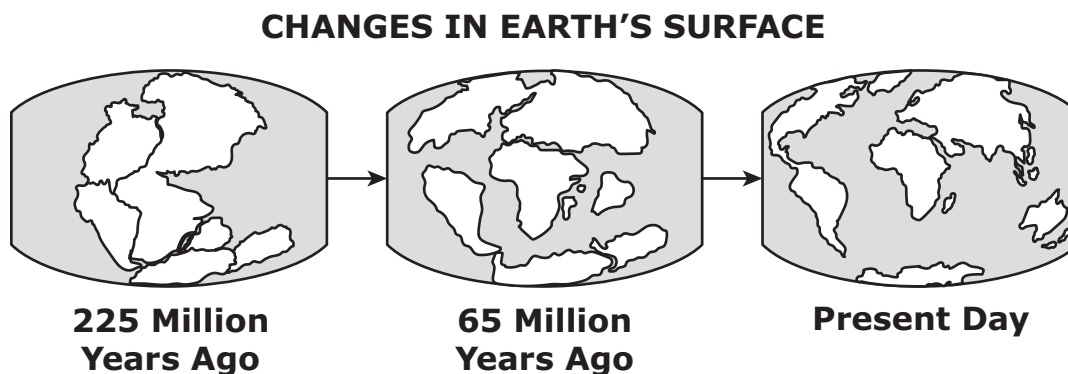
Percentage of Students Answering Correctly: 77%

Reporting Category: Physical Science

Benchmark: SC.6.P.12.1

Benchmark Description: Measure and graph distance versus time for an object moving at a constant speed. Interpret this relationship.

42. Scientists have collected large amounts of evidence that support the idea shown in the illustration below.



Which of the following is a scientific theory that is **best** supported by this evidence?

- Ⓐ evolution
- Ⓑ superposition
- Ⓒ plate tectonics
- Ⓓ universal gravitation

Answer Key: C

Percentage of Students Answering Correctly: 66%

Reporting Category: The Nature of Science

Benchmark: SC.7.N.3.1

Benchmark Description: Recognize and explain the difference between theories and laws and give several examples of scientific theories and the evidence that supports them.

43. A scientist observes six lady beetles of the same species that live in the same location. Each of the lady beetles is a different shade of orange or yellow, and each has a different number of black spots. Which statement **most likely** explains how these differences relate to the evolution of the lady beetle species?

- Ⓐ These differences show that the species may learn to adapt to a warmer and drier climate.
- Ⓑ These differences show that damage to plants in the habitat has caused the species to develop additional spots.
- Ⓒ These differences show that mutations have developed in the population as the species was exposed to pollutants.
- Ⓓ These differences show that there is genetic variation in the population that may allow the species to survive changes to its habitat.

Answer Key: D

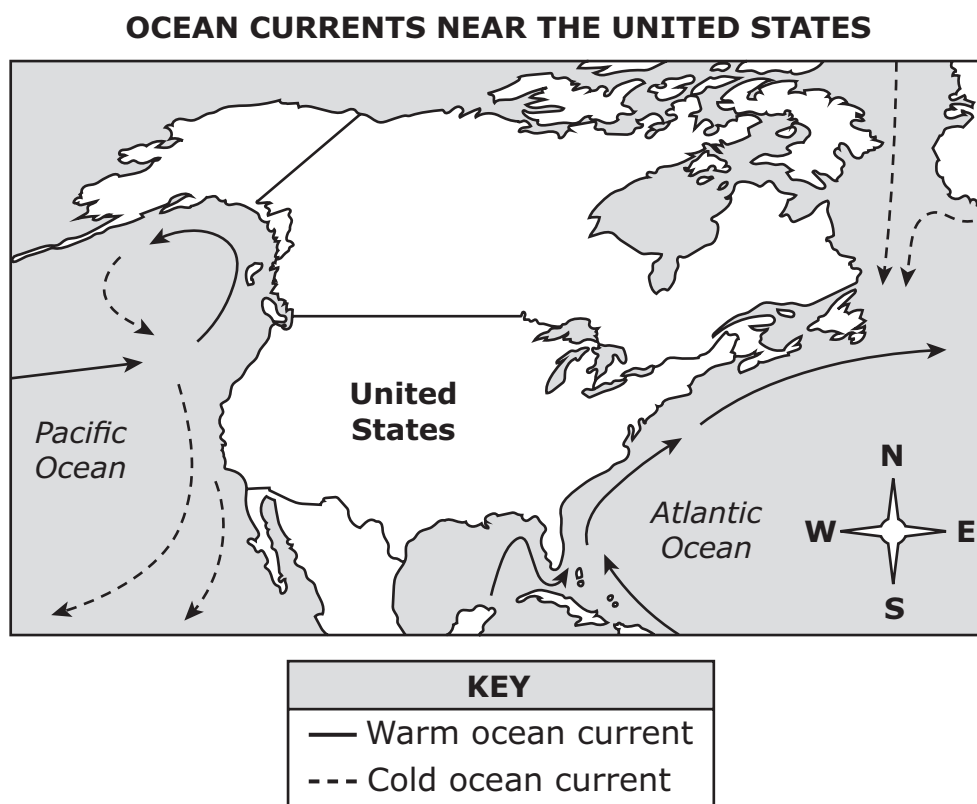
Percentage of Students Answering Correctly: 70%

Reporting Category: Life Science

Benchmark: SC.7.L.15.2

Benchmark Description: Explore the scientific theory of evolution by recognizing and explaining ways in which genetic variation and environmental factors contribute to evolution by natural selection and diversity of organisms.

44. Ocean currents in the Pacific Ocean and the Atlantic Ocean affect climate and weather patterns in the United States. The map below shows some of these ocean currents.



A student researches climate data and determines that the eastern United States usually has higher humidity than the western United States does. Based on the ocean currents shown, which statement **best** explains the difference in humidity?

- Ⓐ The air mass in the Atlantic Ocean is warmer than the air mass in the Pacific Ocean is, and a warmer air mass holds more water vapor than a colder air mass does.
- Ⓑ The air mass in the Pacific Ocean is colder than the air mass in the Atlantic Ocean is, and a colder air mass causes more evaporation than a warmer air mass causes.
- Ⓒ The cold ocean current in the Pacific Ocean causes stronger winds than the warm ocean current in the Atlantic Ocean causes, and moisture evaporates from the Pacific Ocean air mass.
- Ⓓ The warm ocean current in the Atlantic Ocean produces denser air masses than the cold ocean current in the Pacific Ocean produces, and the dense air mass in the Atlantic Ocean cannot release precipitation.

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Answer Key: A

Percentage of Students Answering Correctly: 52%

Reporting Category: Earth and Space Science

Benchmark: SC.6.E.7.3

Benchmark Description: Describe how global patterns such as the jet stream and ocean currents influence local weather in measurable terms such as temperature, air pressure, wind direction and speed, and humidity and precipitation.

- 45.** Dionne observes her science teacher testing a sample of an unknown substance. She recorded her observations in the table shown below.

Test	Characteristic Observed	Yes	No
1	Melts when heated		X
2	Breaks into smaller pieces	X	
3	Dissolves in water		X
4	Reacts with vinegar	X	

Which test would provide evidence of a chemical change?

- Ⓐ 1
- Ⓑ 2
- Ⓒ 3
- Ⓓ 4

Answer Key: D

Percentage of Students Answering Correctly: 71%

Reporting Category: Physical Science

Benchmark: SC.8.P.9.2

Benchmark Description: Differentiate between physical changes and chemical changes.